Women's Education and Health Behaviors: The Mediating Role of Time and Risk Preferences

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Abstract

We examine whether time and risk preferences explain how education affects women's health behaviors in Indonesia. Using waves 4 and 5 of the Indonesian Family Life Survey, we instrument schooling with exposure to the 1970s primary-school construction program and apply several instrumental-variable mediation methods. We study smoking, contraceptive use, and routine health checkups. Across methods, we find little evidence that time or risk preferences mediate education's effects. Results instead point to household economic circumstances as a potential pathway. Estimates differ across behaviors, survey waves, and specifications, with several effects imprecisely estimated. The findings suggest that education alone may not consistently promote healthier behavior and that economic circumstances deserve closer attention when designing education-based health interventions.

Keywords: Education, Health Behaviors, Individual Preferences, Instrumental Variable, Mediation Analysis

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I. Introduction

Education is consistently associated with better health outcomes, yet the reasons for this relationship remain contested (Cutler & Lleras-Muney 2006; Grossman 2006). Although a large literature documents positive education–health gradients, the underlying mechanism is rarely explored (Grossman 2015), and clarifying it is crucial for designing effective policy interventions, particularly in low- and middle-income countries where both education and health outcomes vary widely (Deaton 2002; Strauss & Thomas 1998).

This study explores whether individual preferences, specifically time and risk preferences, mediate the link between education and health behaviors. Using data from the Indonesian Family Life Survey (IFLS) and exploiting variation in schooling exposure created by the SD Inpres school-construction programme, we examine how education affects three important health behaviors in Indonesia: tobacco use, contraceptive use, and participation in health check-ups.

Previous studies have proposed several explanations for the education–health association. One view emphasises confounding factors such as family background, socio-economic status, or innate traits like patience and self-control (Fuchs 1982; Case & Paxson 2011). Another highlights reverse causality, suggesting that healthier individuals are more likely to attain higher education (Grossman 1972; Michael 1973). A third perspective supports a causal interpretation in which education improves health by enhancing decision-making, increasing health knowledge, and improving responsiveness to medical advice (Goldman & Smith 2002; Spandorfer et al. 1995). Evidence from natural experiments and richer survey data increasingly supports this causal view, yet the mechanisms remain unclear.

Time and risk preferences have previously been hypothesized as a “third variable” that influences both educational attainment and health behaviors simultaneously (van der Pol,

2011). However, recent evidence suggests that these preferences are not fixed traits but can be shaped by educational experiences (Perez-Arce, 2017; Jung et al., 2021; Brown & Taylor, 2006; Chong & Martinez, 2021; Tran & Tran, 2024). This evolving understanding opens the possibility that time and risk preferences may act as mediating mechanisms through which education affects health outcomes.

Our analysis focuses on women, reflecting both empirical and policy considerations. Empirically, earlier work finds clearer SD Inpres impacts on women's education than on men's (Akresh et al. 2023; Jung et al. 2021; Mazumder et al. 2019). From a policy perspective, women play a central role in reproductive health and household decision-making, and investing in their education and health yields broad developmental benefits (Onarheim et al. 2016).

The three targeted behaviors have major public-health implications. Indonesia has one of the world's highest smoking prevalences, with 65.5 percent of men and 3.3 percent of women identified as smokers in the 2021 Global Adult Tobacco Survey (WHO 2024). Contraceptive use is a growing concern amid rising HIV prevalence, spurring efforts to improve access to reproductive-health services and promote condom use (Juliastuti et al. 2019). Preventive care is also gaining prominence, as illustrated by the Ministry of Health's recent 183-million-dollar initiative to expand free screenings for non-communicable diseases (Cabinet Secretariat of Indonesia, 2025).

II. Literature Review

Van Der Pol (2011) was among the first to investigate the mediating role of time preference in the relationship between education and health using stated preference to elicit time preference. Including time preference in the regression analysis reduced the coefficient on education, suggesting that time preference may partly explain the impact of education on health. However, these reductions were relatively minor. Importantly, even after adjusting for time preference, the education coefficient remained statistically significant, underscoring a substantial exogenous effect of education on health outcomes.

Grossman (2015) provided a critical review of studies employing instrumental variables to understand the causality between education and health outcomes. The study found that most instrumental variable used have either limited power, does not fulfill the exogeneity criteria, or potentially both. Grossman concluded that the question of whether increased schooling leads to better health outcomes remains open and requires further investigation.

Hamad et al. (2018) conducted a review of existing studies that leverage variations in educational policies to derive more accurate estimates of the health effects of education. They concluded that educational attainment generally has beneficial effects on health across various outcome categories, though the results show significant heterogeneity. Following this, Galama et al. (2018) conducted a review of the literature concerning the impact of education on health outcomes, with a particular focus on the causal effects of education on mortality and its two prevalent preventable causes: smoking and obesity. Their review concluded that education appears to reduce mortality rates among men in certain populations and during specific time periods. However, they found no compelling evidence that education has a direct causal effect on obesity and smoking behavior.

In Indonesia, the research between education and health behaviors is still limited. Several available studies found similar results of the inverse relationship between education

and tobacco consumption. Efendi et al. (2019) found that more than half of the young male population in rural areas were smokers, with significant factors associated with smoking including age, working status, low education level, and exposure to media such as radio and magazines, suggesting a higher likelihood of smoking among those with lower educational attainments. Kusumawardani et al. (2018) indicated that the prevalence of adolescent smoking in Indonesia is significantly affected by education level, among other factors, pointing towards the impact of socio-economic and demographic correlates on cigarette smoking among Indonesian adolescents.

Two studies have directly examined the health impacts of the INPRES school construction program. Mazumder et al. (2021) assessed the long-term and intergenerational effects of SD INPRES, with particular attention to objective health measures such as body mass index (BMI) and blood pressure. Using the Indonesian Family Life Survey (IFLS) and IFLS-East, they found that SD INPRES exposure was associated with a small positive effect on BMI for men, statistically significant at the 10 percent level, but not for women. The program also had a small negative effect on high blood pressure, although this effect was statistically insignificant for both sexes.

Rizal et al. (2023) further explored the impact of the INPRES program on risk factors for non-communicable diseases, using data from the Indonesia Basic Health Research Survey (Riset Kesehatan Dasar, Riskesdas) and the Indonesia National Socio-economic Survey (Survei Sosioekonomi Nasional, Susenas). They found that exposure to an additional primary school per 1,000 children increased the probability that women would have a BMI of 25 kg/m^2 or higher by 1.6 percentage points. No evidence was found of an effect on the weight status of men or on blood pressure for either sex. Additionally, the study reported a small negative effect of INPRES exposure on the probability of being a current smoker.

The only study identified to have conducted a quasi-mediation analysis on the link between education and health behavior in Indonesia is Jung et al. (2021). Using data from the fifth wave of the Indonesian Family Life Survey (IFLS-5), the study assessed the effect of education on time preference by exploiting variation in exposure to the Indonesian INPRES primary school construction program of the 1970s as an instrument. The results showed that each additional year of education reduced the likelihood of a female respondent selecting the most impatient option by 9 percent. Although the study did not directly examine the subsequent impact on health behaviors, it reported that a lower discount rate was associated with greater contraceptive use among women and lower smoking prevalence among men.

III. Methodology

Understanding the causal impact of education on health behaviors requires addressing the non-random assignment of educational attainment. If education was randomly assigned, one could straightforwardly estimate its causal effect on health behavior using Ordinary Least Squares (OLS) regression. However, the potential for endogeneity necessitates a more robust approach. This study adopts an instrumental variable (IV) approach to ascertain the causal relationship.

The IV chosen for this analysis is the Indonesian school construction program (SD Inpres), following the seminal paper by Duflo (2001). The SD Inpres program was launched in the early 1970s to increase access to education in Indonesia, primarily through the construction of primary schools throughout the country. The program's allocation was based on regions' needs rather than individual student characteristics, making it a suitable instrument for education in this context.

To explore how education affects health behaviors, we adopt a mediation strategy following the approach of Maccini and Yang (2009). In this framework, the endogenous mediating variable is included together with the primary independent variable. If adding the mediating variable significantly reduces the association between education and the health behavior outcome, and the mediating variable is strongly correlated with the outcome itself, this would suggest that the mediating variable captures part of the influence of education on health behavior.

Since the outcome variable in our models are all binary variable, instead of using the normal IV regression, we utilize the IV-Probit model. The IV-Probit will use the following specification:

$$Pr_i(y = 1|X, Z) = X\beta + \widehat{E_1}\gamma + \epsilon \quad (1)$$

The year of education is modeled using the instrument Z_i:

$$E_i = X\beta + Z_i\delta + \epsilon \quad (2)$$

Where:

Y_i = Binary outcome variable (health behaviors)

X = Vector of Exogenous covariates

E_i = Endogenous variable (years of education)

In this study, we adopt the same set of control variables as Duflo (2001), including the interaction between year of birth and the proportion of the population aged 5 to 14 in 1971, the district-level school enrollment rate in 1971, and the allocation of the water and sanitation program, which was the second-largest INPRES initiative at the time. We further control for individual characteristics, namely religious affiliation (Islam), Javanese ethnicity, primary language spoken, age, urban-rural, chronic disease, and number of dependencies.

This approach helps address the endogeneity of education while allowing for non-linear regression of the outcome. However, it does not provide a quantitative estimate of the relationships between the variables and should only be taken as suggestive.

To obtain a quantitative estimate, we apply the mediation analysis framework developed by Dippel et al. (2020), known as the ivmediate model. This method allows us to identify both the total treatment effect and the mediated effect using only a single instrument, provided that the treatment and the mediator are influenced by the same unobserved confounders. In our context, this assumption is plausible because time preferences, risk preferences, and health behaviors are shaped by common underlying factors such as cognitive ability, socio-economic background, and personality traits. These same factors are also known to affect educational attainment. Therefore, we argue that this is appropriate.

The mediation analysis allows us to parse out the direct effects of education on health behaviors from the indirect effects through the mediators. We first estimate the following equations to find the effect of education on mediating variables:

$$\text{First Stage: } E_i = \beta_E^Z \times Z_i + \epsilon_i \quad (3)$$

$$\text{Second Stage: } M_i = \beta_M^E \times \widehat{E_i} + \omega_i \quad (4)$$

The final effect of education on health behaviors, the mediating parameter, and the direct effect of the treatment are identified by the following two stages least squares:

$$\text{First Stage: } M_i = \gamma_m^Z \times Z_i + \gamma_M^T \times E_i + \xi_i \quad (5)$$

$$\text{Second Stage: } Y_i = \beta_M^Y \times \widehat{M_i} + \beta_y^E \times E_i + \eta_i \quad (6)$$

Where:

E_i : educational attainment of individual i,

Z_i: instrument capturing the exposure to the SD Inpres program for individual i

Y_i : health behaviors of individual i

The direct estimation of TE can be obtained by substituting equation (3) into (6):

$$\begin{aligned} Y_i &= \beta_Y^M \times (\beta_M^E \times E_i + \epsilon_M) + \beta_Y^E \times E_i + \epsilon_Y \\ &= (\beta_Y^M \times \beta_M^E + \beta_Y^E) \times E_i + \beta_Y^M \epsilon_M + \epsilon_y \end{aligned} \quad (7)$$

Here, the $(\beta_Y^M \times \beta_M^E + \beta_Y^E)$ represents the total effect of education on health outcomes. However, this approach requires the use of a linear probability model (LPM) in the second stage, which introduces several limitations. A known issue with LPM is that it may generate predicted values outside the [0,1] range. Nevertheless, this is not a major concern when the objective is to estimate average marginal effects rather than to predict individual probabilities. A more substantive limitation is the underlying assumption of linearity in the relationship between education and health behaviors, which is unlikely to be satisfied in this context and may result in specification bias.

To complement the IV-based mediation approach, we employ Stata's causal mediation analysis using the *mediate* command. This method allows us to express the general form as an integral of the conditional expectation of the outcome with respect to the conditional distribution of the mediator (Luedicke, 2023). Because we are still unsure about the mediating effect, we follow the decomposition proposed by Nguyen, Schmid, and Stuart (2021), which

separates the total effect into the natural direct effect (NDE) and the natural indirect effect (NIE). The **natural direct effect (NDE)** captures the average effect of the treatment when the mediator is held fixed at the level it would take under no treatment, $Y[1, M(0)] - Y[0, M(0)]$. In contrast, the **natural indirect effect (NIE)** represents the portion of the treatment effect that operates through the mediator, holding the treatment status constant, and is defined as $Y[1, M(1)] - Y[1, M(0)]$. This decomposition framework allows the total effect to be expressed as the sum of these two components. To estimate these effects, we express the outcome in terms of an integral over the conditional distribution of the mediator, following Luedicke (2023):

$$E\big[Y_i\big(e, M_i(e')\big)\big|X_i = x\big] = \int E[Y_i|M_i = m, E_i = e, X_i = x]dF[m|E_i = e', X_i = x] \quad (8)$$

This approach allows us to combine linear and non-linear model in the mediation analysis. However, because the *mediate* command does not support the use of instrumental variables within its framework, we utilize the control function method. This method addresses endogeneity by first estimating the endogenous treatment variable using an instrument in a linear regression. From this first stage, we obtain the residuals ν_i, which capture the endogenous component of the treatment that is correlated with unobserved factors affecting the outcome (Wooldridge, 2015). These residuals are then included as an additional covariate in the mediation analysis to adjust for the endogeneity of the treatment variable.

Combining control function method with the *mediate* command gives us the following econometrics specification:

$$\big[Y_i\big(e, M_i(e')\big)\big|X_i = x, \nu_i\big] = \int E[Y_i|M_i = m, E_i = e, X_i = x, \nu_i]dF[m|E_i = e', X_i = x, \nu_i] \quad (9)$$

One downside of this approach is that the standard errors in the second stage may be underestimated because the residual is a generated regressor, constructed from the first-stage

estimation rather than directly observed. To address this issue, we employ the Jackknife method for standard error estimation. The Jackknife approach is used over bootstrap due to our limited number of observations and the high number of interaction terms in the equation. Using bootstrap significantly reduces the number of successful repetitions, leading to widely varying estimates across iterations. Additionally, the Jackknife method has the unexpected advantage of overestimating standard errors, which may help counteract the underestimation issue (Pischke, 2007).

In our analysis, because we are considering multiple hypothesis tests simultaneously, standard statistical techniques will lead to over-rejection of null hypotheses unless the multiplicity of the testing framework is explicitly considered. Here we use the Romano-Wolf multiple hypothesis correction which controls the familywise error rate (Clarke et al., 2019).

IV. Data and Sample

This study utilizes data from the fourth wave of the Indonesian Family Life Survey (IFLS-4), collected during 2007–2008, and the fifth wave (IFLS-5), collected during 2014- 2015. The analysis relies on time preference and risk preference measures obtained from two sets of hypothetical scenarios, referred to as Module A and Module B.

Figure 4.1 summarizes the time preference choices in Module A, where participants weigh a certain immediate payment against various larger amounts available after one year. This design captures preferences over shorter horizons, with relatively modest increments in the delayed rewards. In contrast, Figure 4.2 presents the decisions from Module B, which stretches the waiting period to five years, offering much larger increases in future payments.

Addressing the validity and differences between these two modules is crucial. Previous research suggests that people tend to be more impatient about rewards in the near future than those in the distant future (Yue & Wang, 2017). Common approaches to time discounting often assume that the same nominal amount of money provides identical utility at any point in time. However, as Frederick et al. (2002) have emphasized, inflation causes individuals to devalue future payments, which effectively raises discount rates as the delay increases. In addition, the possibility of intertemporal arbitrage, where one chooses a smaller immediate amount to invest with the aim of outperforming the larger delayed sum, adds further complexity to these estimates. Both inflation-related concerns and arbitrage opportunities hinge on the respondent's financial literacy, which strongly correlates with education (Lusardi, 2019). This force us to interpret the results obtained from time preference measure B with much caution

To ensure a consistent metric for comparing Modules A and B, we follow the methodology employed by Jung et al. (2021) to construct an "imputed" time preference measure. Under the assumption of exponential discounting, if a participant prefers IDR 1 million now over IDR 3 million in one year, then their discount factor (β) must be no greater

than 1/3. Conversely, if they choose the delayed IDR 3 million, their β must be at least 1/3. For someone who picks IDR 1 million now over IDR 3 million delayed by one year, but would switch to IDR 6 million after one year instead of taking IDR 1 million now, we can narrow their β to between 1/6 and 1/3. We then take the midpoint of these bounds and invert it, so that higher values indicate greater impatience. If an upper bound is not available, we use the lower bound alone to generate a conservative estimate of the discount rate. For Module B, which involves a five-year delay, we adjust this measure to the fifth-root power to reflect the longer horizon.

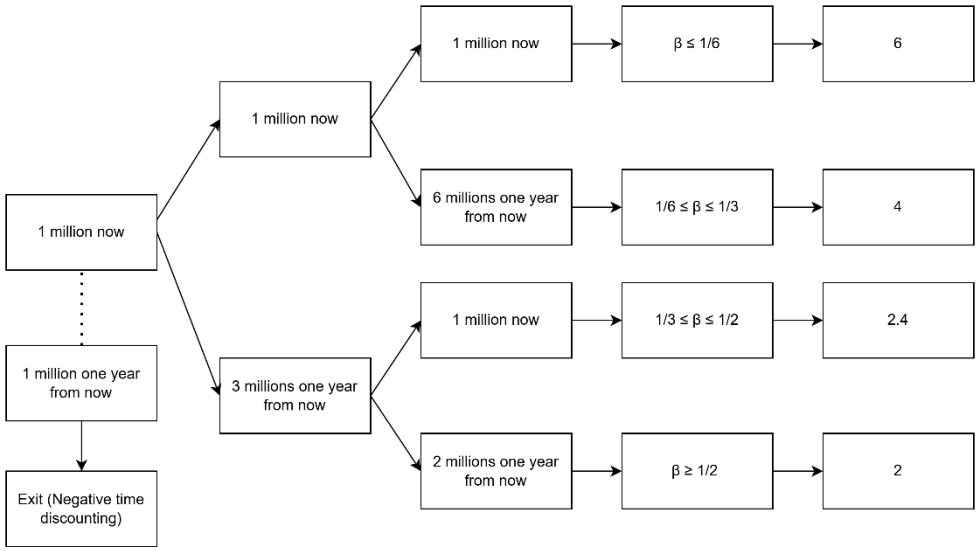


Figure 4.1 Time Preference Module A

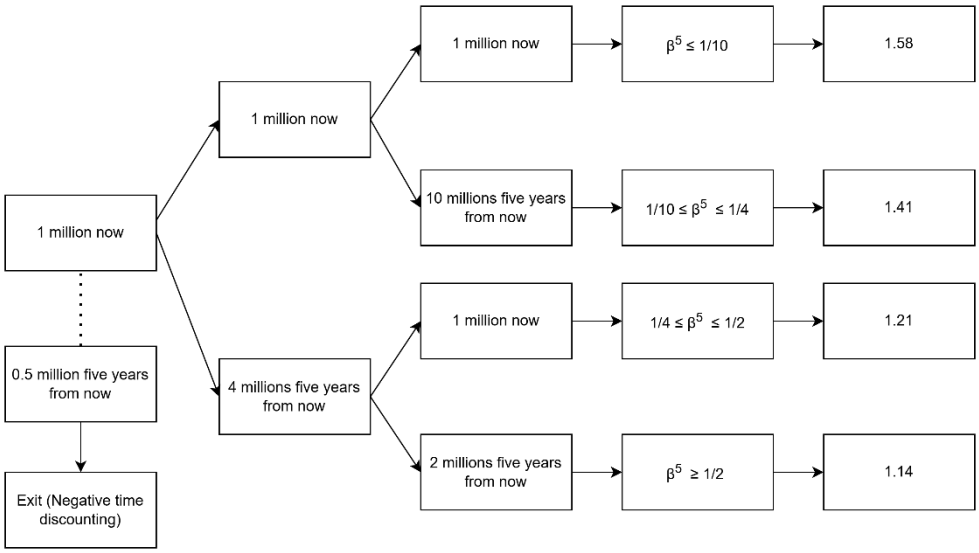


Figure 4.2 Time Preference Module B

For our risk preference analysis, we employ the imputed risk preference measure introduced by Kimball et al. (2008). Let $U(W)$ represent an individual's concave utility function defined over their original lifetime income W. Now consider a 50-50 gamble in which lifetime income is either multiplied by a factor γ (with $\gamma > 1$) or reduced by a fraction π. The individual will accept this risky job if its expected utility is greater than the utility associated with a sure outcome. Formally, this condition can be expressed as:

$$0.5(U(\gamma \mathrm{W}) + 0.5(\mathrm{U}((1-\pi)\mathrm{W}) \geq \mathrm{U}(\mathrm{W})$$

We assume that constant relative risk aversion (CRRA) well approximates individuals' utility over the income. The equation used is as follows

$$U(W) = \frac{W^{1-\frac{1}{\theta}}}{1-\frac{1}{\theta}}$$

Where θ is the relative risk tolerance. The W is the payoff, normalized to 1 based on the module (IDR 800,000 in the first module and IDR 4,000,000 in the second module). We then derive the bounds of risk tolerance for each respondent. The midpoint of these bounds is calculated for each respondent and is illustrated in Figures 4.3 and 4.4. For choices without any

midpoint, we take the conservative estimate and use the bounds as the risk preference parameter.

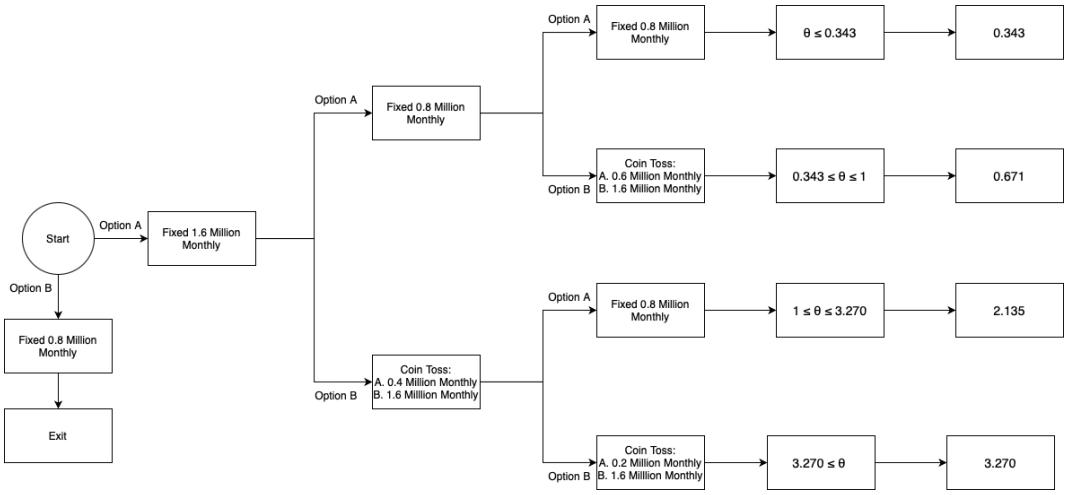


Figure 4.3 Risk Preference Module A

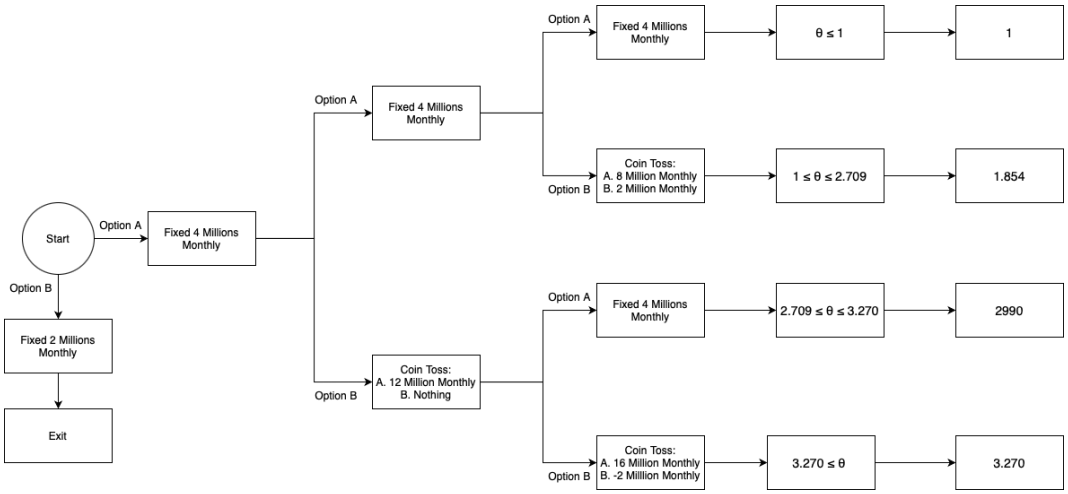


Figure 4.4 Risk Preference Module B

Assuming a concave utility function, it follows that individuals will display increasing risk aversion when faced with larger stakes, such as those presented in Module B. This is because the concavity of the utility function implies diminishing marginal utility of wealth, making large, uncertain gains less appealing relative to safer outcomes. Additionally, the framing of the choices plays a significant role: the final option in Module B includes a potentially negative payoff, which, according to Prospect Theory (Kahneman & Tversky, 1979), heightens loss aversion and thus strengthens the tendency to avoid risk. Together, these

factors suggest that respondents are more likely to be risk-averse in Module B compared to Module A.

For the health behaviors, we rely on three commonly examined outcomes: contraceptive use, smoking habits, and whether the respondent had a health checkup within the last four months. Each of these behaviors has been empirically associated with time preference (Bickel et al., 1999; DellaVigna & Paserman, 2005; Do & Shin, 2017). In this analysis, the smoking variable is binary: it equals 1 if the respondent is actively smoking at the time of the survey and 0 if she is not. Similarly, contraceptive use is measured as a binary variable, taking the value of 1 if either the respondent or her husband reported using contraceptive at the time of the survey, and 0 otherwise.

V. Results

5.1 Descriptive Statistics

Table 5.1 presents the summary statistics for the time preference module. In both modules, most respondents prefer the immediate reward, with this tendency being even more pronounced in Module B.

Table 5.1 Summary Statistics of Time Preference

	IFLS-4 (N = 1,057)	IFLS-5 (N = 1,144)
Time Preference A		
1 (Most Patient)	0.10	0.11
2	0.07	0.09
3	0.17	0.18
4 (Most Impatient)	0.66	0.62
Time Preference B		
1 (Most Patient)	0.02	0.02
2	0.04	0.06
3	0.16	0.18
4 (Most Impatient)	0.78	0.74

The risk preference as shown in Table 5.2 exhibits a similar pattern: most individuals choose the most risk-averse option. However, this preference is less pronounced in Module A. This follows from the underlying difference between the two measures explained previously.

Table 5.2 Summary Statistics of *Risk Preference*

	IFLS-4 (N = 1,057)	IFLS-5 (N = 1,144)
Risk Preference A		
1 (Most Risk Seeking)	0.22	0.23
2	0.14	0.19
3	0.08	0.16
4 (Most Risk Averse)	0.56	0.42
Risk Preference B		
1 (Most Risk Seeking)	0.07	0.05
2	0.03	0.04
3	0.10	0.12
4 (Most Risk Averse)	0.80	0.79

Table 5.3 reports the correlation matrix for the pooled sample. There is a high degree of correlation between the two risk preference modules and the two risk preference modules. However, the correlation between time preference and risk preference is low and statistically insignificant, which is a surprising result given that these variables are often modelled together in the literature (Cohen et al., 2020).

Table 5.3 Correlation Matrix for Risk and Time Preference

IFLS4

		Time Preference		Risk Preference	
		A	B	A	B
Time Preference	A	1.00	0.59	0.02	0.01
	B		1.00	0.05	0.04
Risk Preference	A			1.00	0.44
	B				1.00

IFLS5

		Time Preference		Risk Preference	
		A	B	A	B
Time Preference	A	1.00	0.52	0.07	0.12
	B		1.00	0.12	0.18
Risk Preference	A			1.00	0.25
	B				1.00

Figure 5.1 shows the distribution of education levels in both IFLS-4 and IFLS-5. As seen from the figure, the distribution in both waves are nearly identical, which is expected given that we use the same cohort across waves and achieve a re-interview rate of 48% in our sample. Approximately 40% of the sample completed only primary education (SD) or less. Around 25% progressed to junior secondary school (SMP), and 20% completed senior secondary education (SMA). In contrast, only about 10% attained a university degree (S1)

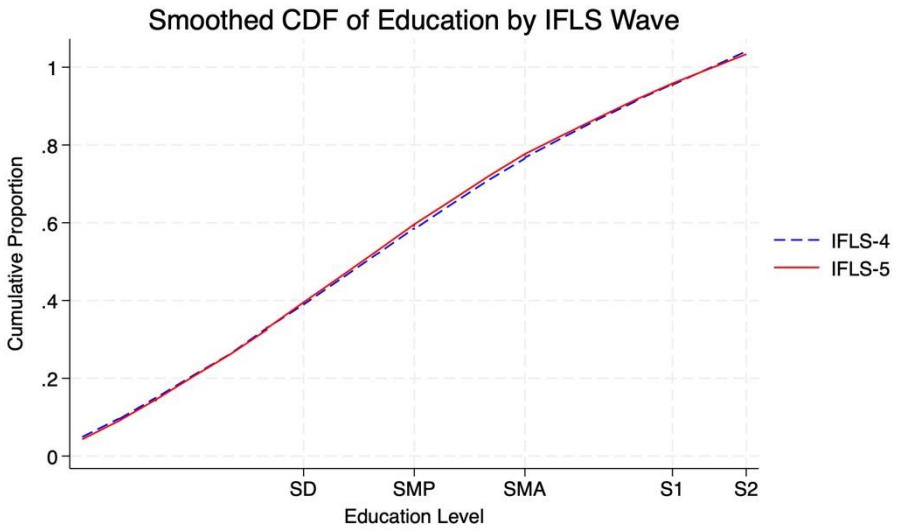


Figure 5.1 Education CDF

Figure 5.2 provides a locally weighted scatterplot smoothing (LOWESS) plot of education and imputed time preference. The lines for IFLS-4 and IFLS-5 nearly overlap, indicating a stable relationship over time. Notably, Modules A and B show different patterns: as years of education increase, time preference decreases in Module A but increases in Module B.

Hyperbolic discounting suggests that as delays lengthen, individuals should become more patient, with the rate of discounting declining over time (Laibson, 1997; Frederick et al., 2002). This implies that extending the delay from 1 year (Module A) to 5 years (Module B) should lead to lower discounting. However, our findings contradict this prediction: more educated individuals exhibit greater impatience in Module B than in Module A, favoring immediate rewards more strongly over the longer horizon.

This deviation can be explained by opportunity costs and future income expectations. More educated individuals are more financially literate and aware of alternative investment opportunities, making them more likely to discount long-term rewards more aggressively (Becker & Mulligan, 1997; Brown et al., 2016). Over a 5-year period, they may anticipate higher returns from investing the immediate reward rather than waiting for the delayed

payment, leading to increased impatience. In contrast, a 1-year delay offers fewer investment opportunities, so their discounting behavior in Module A aligns more with traditional models (Epper et al., 2011).

Additionally, future income expectations further explain this pattern. More educated individuals anticipate rising earnings over time, making a future lump sum relatively less valuable compared to their expected financial position (Strotz, 1955; Modigliani & Brumberg, 1954). While less educated individuals may view the delayed reward as a significant financial gain, highly educated individuals discount it more heavily because they expect to be in a stronger financial position in five years, reducing the perceived benefit of waiting (Haushofer & Fehr, 2014).

In this paper, we do not aim to explore the various models of time preference in detail. However, this statistic strengthens our confidence in applying the imputation method for time preference, as it aligns with the assumption of exponential discounting, while our findings explicitly contradict the predictions of the hyperbolic discounting model.

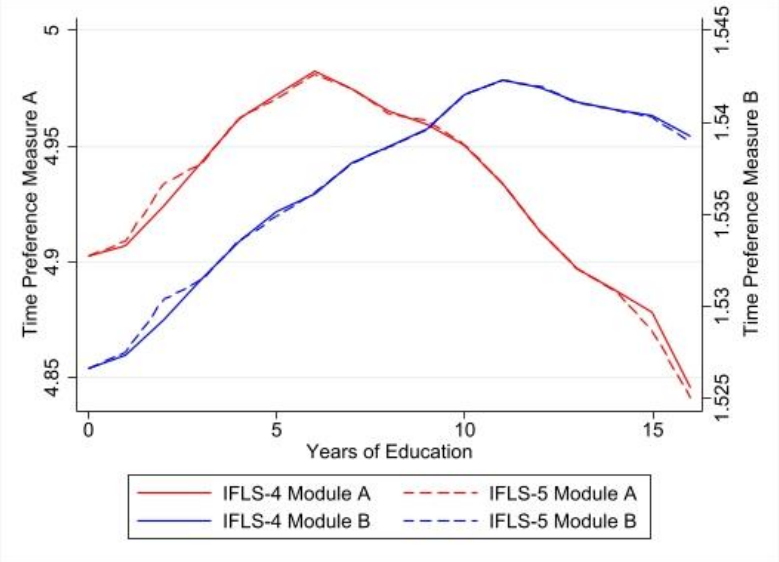


Figure 5.2 Education and Time Preference LOWESS Plot

Figure 5.3 shows that risk preference scores decline with additional schooling through the primary and junior-high years, reach a minimum near the end of junior high, and then rise over senior-high and university levels, producing a U-shaped relationship. This pattern is present in Module A for both IFLS waves and in Module B for IFLS-5, whereas Module B for IFLS-4 displays only a slight early decline followed by a modest increase. Although the two waves do not align as closely as in the risk preference graph, the overall shape remains similar across waves in Module A and diverges chiefly in Module B.

Modules A and B both aim to measure risk preference, but they differ in their framing and potential biases. Module A presents a more gradual increase in risk-seeking behavior, capturing a broad and incremental shift in attitudes toward uncertainty. Module B, on the other hand, introduces higher stakes, with the most risk-seeking option involving a potential negative outcome of -2 million monthly, which is close to the average salary in many contexts. This design creates a potential confounding effect, as lower-income individuals may avoid the highest-risk option not because they are inherently less risk-seeking, but because they cannot afford the financial loss (Barsky et al., 1997; Chiappori & Paiella, 2011). In contrast, wealthier individuals may appear more risk-seeking simply because they have the financial flexibility to absorb potential losses, rather than because they have a stronger preference for risk (Dohmen et al., 2011).

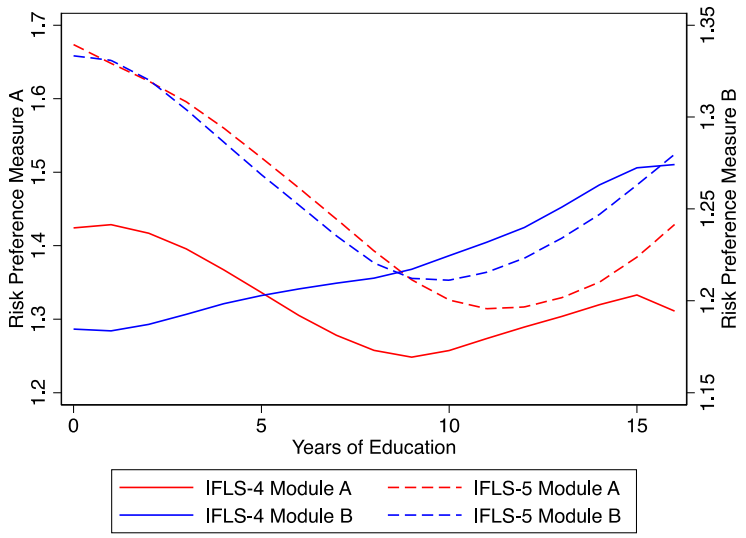


Figure 5.3 Education and Risk Preference LOWESS Plot

Table 5.4 summarizes the share of respondents who smoke, use contraceptive, and have had a health checkup in the last four months. Smoking remains quite rare in both survey waves, with only about 4–5% of participants reported smoking. As a result, any conclusions about smoking habits should be drawn carefully, given the small number of cases and the increased chance of statistical noise. Contraceptive use and regular health checkups also appear uncommon, suggesting that preventive health behaviors and services are still not widespread.

Table 5.4 Summary Statistics for Health Behaviors

		IFLS-4	IFLS-5
Smoking Behavior	Yes	0.04	0.05
	No	0.96	0.95
Contraceptive Use	Yes	0.10	0.09
	No	0.90	0.91
Health Checkup	Yes	0.15	0.08
	No	0.85	0.92

Table 5.5 presents the correlations between time preference and risk preference measures and various health behaviors. The correlations are generally low, and there is a lack of consistency in the direction and magnitude of these relationships between the IFLS-4 and IFLS-5 samples.

Table 5.5 Correlation Matrix for Preferences and Health Behaviors

		Time Preference Module A	Time Preference Module B	Risk Preference Module A	Risk Preference Module B
IFLS-4	Smoking Behavior	-0.072	-0.057	-0.097	0.021
	Contraceptive Use	0.015	0.017	0.013	-0.008
	Health Checkup	-0.038	-0.041	0.035	0.050
IFLS-5	Smoking Behavior	0.012	0.002	-0.014	0.012
	Contraceptive Use	-0.049	0.007	-0.010	-0.047
	Health Checkup	0.019	0.002	-0.009	-0.045

Figure 5.4-5.6 illustrates the relationship between education and various health behaviors across IFLS-4 and IFLS-5. For both contraceptive use and health checkups, the patterns appear consistent across the two waves. Contraceptive use follows an inverted U-shape, peaking at intermediate levels of education, while health checkup rates increase steadily with higher education levels. In contrast, smoking behavior shows differing trends. In IFLS-4, the relationship appears relatively flat, with little variation across education levels, whereas in IFLS-5, smoking behavior declines as education increases. While these visual patterns offer an initial sense of the relationships, they should be interpreted with caution, as the graphs do not account for sample size variation or statistical significance, and the observed associations may be relatively weak.

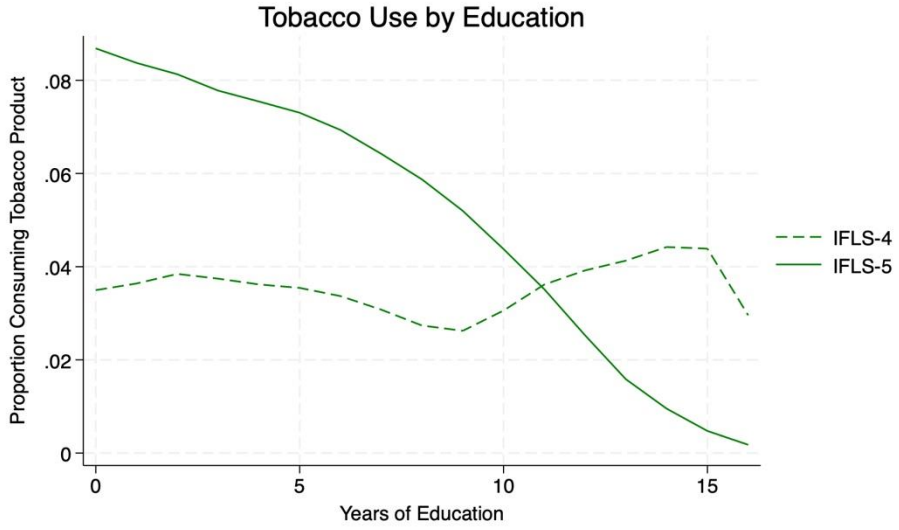


Figure 5.4 Education and Smoking Behavior

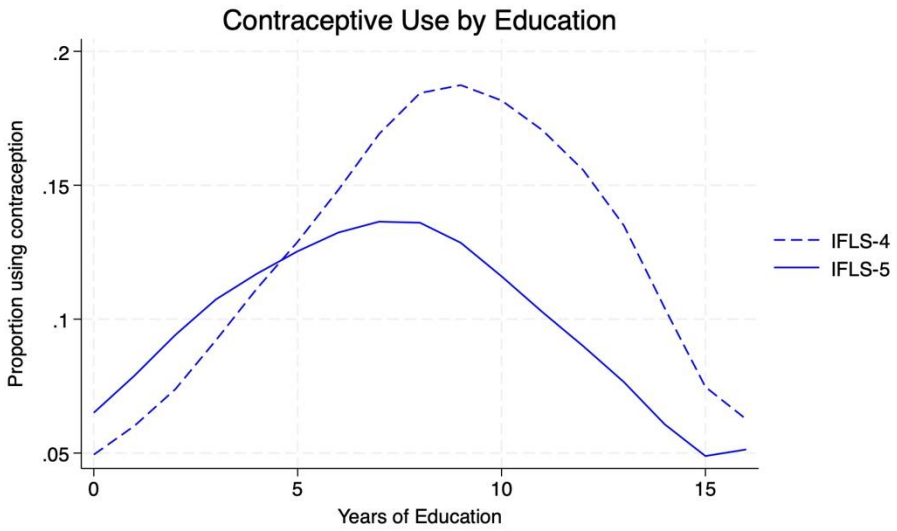


Figure 5.5 Education and Contraceptive Use

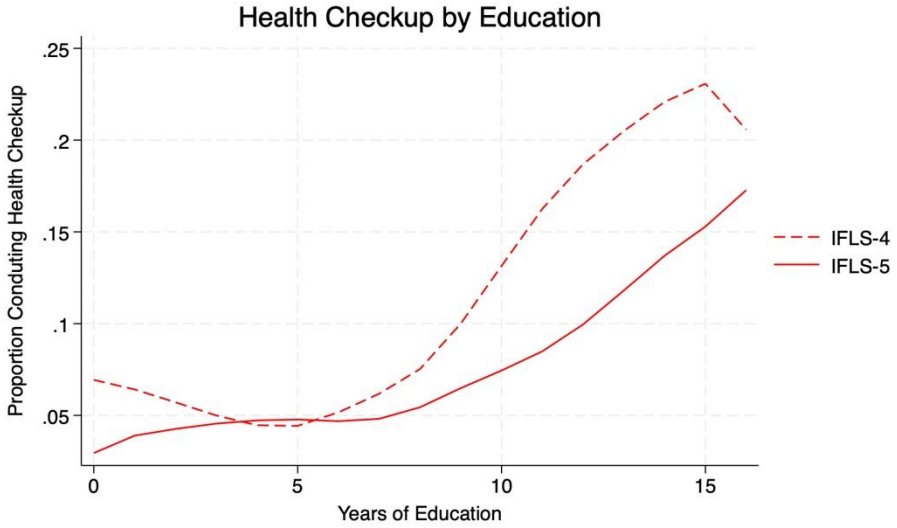


Figure 5.6 Education and Health Checkups

5.2 Education and Individual Preferences

We first check for the strength of the instrumental variable. Table 5.6 reports the first-stage results from the IV-2SLS estimation. The coefficient on the interaction of INPRES intensity with the treated cohort dummy is significant across all sample groups, aligning with earlier studies using IFLS data that found effects of SD INPRES construction (Duflo, 2004; Jung et al., 2021; Mazumder et al., 2019). Although the F-statistic is lower than the conventional rule-

of-thumb threshold of 10, recent work by Angrist & Kolesár (2024) indicates that as long as the first-stage coefficient is significant and endogeneity is not extreme, the second-stage t-tests remain valid even when the first stage is weak.

Table 5.6 SD Inpres on Years of Education

	Years of Education	
	IFLS-4	IFLS-5
INPRES Intensity * Treated Cohort Dummy	0.528** (0.225)	0.513** (0.212)
Observations	1,057	1,114
F-Statistics	5.587	5.908

Given that the program targeted less developed areas, we want to show that our effects are not explained by general catching up (or mean reversion) in those areas, as this would violate the parallel trends assumption. To do this, we follow the procedure from Akresh et al. (2023) by estimating placebo regressions that compare the old cohort (ages 12–17 in 1974) with an even older cohort (ages 18–24 in 1974). The result is shown in Appendix 1. The placebo regressions show no statistically significant effects, which provides suggestive evidence that the main difference-in-differences results are not driven by a failure of the identification assumption.

After validating the instrumental variable, we estimate the effect of education on individual preferences using the control-function approach rather than the conventional two-stage least squares procedure. The control-function method allows us to accommodate the potential non-linear relationship between education and preferences suggested by Figures 5.2 and 5.3. Complete estimation results are reported in Tables 5.7 and 5.8.

The findings indicate a non-linear pattern: the linear term for years of education is not statistically significant in either survey wave, whereas the squared term is significant in Wave 5. This result contrasts with Jung et al. (2021), who found a significant positive linear effect of years of education on time preference in the Wave 5 sample.

Table 5.7 Years of Education and Individual Preferences IFLS-4

	Time Preference		Risk Preference	
	Module A	Module B	Module A	Module B
Years of Education	-0.206 (0.162)	-0.009 (0.010)	-0.047 (0.131)	0.020 (0.060)
Years of Education Squared	0.001 (0.002)	0.0001 (0.0001)	0.0006 (0.0001)	-0.001 (0.0009)
Observations	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust Standard errors in parentheses. We include controls for ethnicity, religion, age, language used for the interview, and relevant controls from Duflo (2001).

Table 5.8 Years of Education and Individual Preferences IFLS-5

	Time Preference		Risk Preference	
	Module A	Module B	Module A	Module B
Years of Education	0.135 (0.329)	0.017 (0.026)	0.027 (0.254)	-0.117 (0.122)
Years of Education Squared	-0.005*** (0.002)	-0.0004*** (-0.0001)	-0.0009 (0.0016)	-0.002*** (0.0007)
Observations	1,114	1,114	1,114	1,114

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust Standard errors in parentheses. We include controls for ethnicity, religion, age, language used for the interview, and relevant controls from Duflo (2001).

5.3 Instrumental Variable-Probit Analysis

We begin the mediation analysis with smoking behavior (Table 5.9). In the baseline model, higher education significantly increases the likelihood of smoking. When we add expenditure as a proxy for socioeconomic status, the effect of education grows stronger, suggesting that economic level partly mediates the relationship between education and smoking. In IFLS-5, risk preference measure B has significant positive effect on smoking behavior, even when controlling Nevertheless, adding these preference measures does not alter the education coefficient, suggesting that time and risk preferences do not mediate the effect of education on smoking.

Turning to contraceptive use (Table 5.10), we see differences between IFLS-4 and IFLS-5. In IFLS-4, none of the variables is significantly related to contraceptive use. In IFLS-5, however, higher education corresponds to lower contraceptive use, and this effect intensifies when we include expenditure, which itself is positively associated with contraceptive. Risk preference measure B has significant negative relation in the IFLS-5 result, aligning with the idea that more risk-averse individuals are more likely to use contraceptive. Still, the inclusion

of the preferences does not substantially change the impact of education, implying that preferences do not explain the observed relationship between education and contraceptive use.

Finally, Table 5.11 shows that risk preference influences regular health checkups in both IFLS-4 and IFLS-5, as more risk-averse individuals are more likely to have regular checkups. When we control for expenditure, education becomes negatively associated with checkups. This suggests that while higher economic status (partly driven by education) encourages checkups, education may also be linked to other factors that reduce the likelihood of checkups. Risk preference measure B also show a significant negative relationship with probability of health checkup. The introduction of individual preferences does not significantly alter the education coefficient, indicating that these preferences do not account for the negative relationship between education and health checkups.

Appendices 2–4 report the Average Marginal Effects (AMEs) from the IV-Probit estimations. None of the estimated AMEs are statistically significant. This likely reflects the low incidence of the reported health behaviors in the sample, which leads predicted probabilities to concentrate near the extremes (i.e., close to 0 or 1). As marginal effects in nonlinear models are attenuated at these extremes, the resulting AMEs tend to be small in magnitude, reducing the precision and statistical significance of the estimates.

The results indicate that education, expenditure, and risk preference (module B) are strong predictors of all three health behaviors. However, among these factors, only the relationship between education and health behaviors can be interpreted causally.

5.4 *Ivmediate* Analysis

Table 5.12, 5.13, and 5.14 shows the result of the *ivmediate* analysis for smoking behavior, contraceptive use, and health checkup. For smoking behavior, the results across both IFLS-4 and IFLS-5 show that none of the estimated coefficients for total, direct, or indirect effects are statistically significant.

Similarly, for contraceptive use, the total, direct, and indirect effects are all statistically insignificant in both IFLS-4 and IFLS-5. Even after controlling for expenditure levels, the results remain unchanged, suggesting that financial constraints do not drive this relationship. While some coefficients in IFLS-5 are negative, their statistical insignificance prevents any firm conclusions about a possible inverse relationship between education and contraceptive use.

In contrast, the findings for health checkup behavior tell a different story. In IFLS-4, the direct effect of education on health checkup participation is positive and statistically significant across all mediators, with coefficients around 0.018 to 0.019 ($p < 0.01$).

The previous analysis using the standard instrumental variable model (see Table 5.9) indicated that in the IFLS-4 sample, the effect of education on health checkups was positive but not statistically significant. This is similar to the finding from the *ivmediate* analysis which found that the total effect, while positive, is not statistically significant. Below we examine the potential limitations of this analysis:

5.4.1 The Use of Linear Probability Model

The Linear Probability Model assumes constant marginal effects of the explanatory variables, which can limit its usefulness in contexts where the expected effect, such as that of time preference, is minimal, thereby reducing statistical power. In addition, if the treatment effect is heterogeneous, meaning the impact of education on health behaviors varies across individuals, the average effect may cancel out, resulting in a non-significant estimate. As a result, even if education influences health behaviors for specific subgroups, the aggregation of differing effects may obscure a clear overall relationship.

To test for this possible cause, we use non-binary variables in the place of initial variables. For smoking behavior, we instead use weekly tobacco expenditure. The result can be seen in Appendix 5. The table shows that even with a non-binary outcome variable, the

effect sizes are still not statistically significant. This indicates that specification error may not be an issue in the case of tobacco expenditure.

Additionally, a known limitation of the Linear Probability Model is that its predicted values are not confined between 0 and 1. As shown in Appendix 6, the distribution of predicted values includes negative values, which would be problematic if the model were used for prediction, as probabilities outside the [0,1] range are not meaningful. However, as argued by Angrist (2001), this is not necessarily a concern when the model is used to estimate average marginal effects rather than to predict individual outcomes. In our case, since the focus is on identifying the effect of education on health behaviors rather than generating accurate probability predictions, the fact that some predicted values fall outside the [0,1] interval does not compromise the validity of our estimates.

5.4.2 Limited Sample

Individual preferences have consistently demonstrated limited explanatory power in predicting behavior. As a result, a large sample size is necessary to enhance the statistical power of the analysis and reduce the likelihood of false negative. However, the limited sample size in each dataset may constrain the statistical power available in this study, potentially contributing to the lack of statistical significance.

To address this issue, we attempt to increase statistical power by combining observations from IFLS-4 and IFLS-5. For respondents interviewed in both waves, only the most recent observations are used. Additionally, expenditures from IFLS-4 are adjusted for inflation. The results of this approach are presented in Appendix 4. Due to the panel nature of the IFLS data, the total sample size only increases to 1,684 observations. The estimates from the combined dataset closely resemble those obtained from the IFLS-5 sample, with little to no variation in the results.

To be more convincing, we follow the approach used by Jung et al. (2021), which involves classifying the preference variable into a binary indicator. Grouping the variable in this way increases statistical power by simplifying the variation and reducing noise, making it easier to detect meaningful associations. However, this comes at the cost of losing finer distinctions within the original variable, potentially overlooking subtle but relevant variations in preference that may carry explanatory value. The result is shown in Appendices 7-9. While some coefficients did change, none of them are different from the ones using imputed value.

5.4.3 Non-linearity of Preference Measure

In the previous results, we assumed a linear relationship between the time and risk preference measures and health behaviors. However, several studies have argued that time and risk preferences may exhibit non-linear patterns (Andersen et al., 2008; Abdellaoui et al., 2013). For instance, non-linearities may arise due to diminishing sensitivity, probability weighting, or threshold effects that are not captured in a linear framework.

To account for this possibility, we re-estimated our models using the log-transformed values of the imputed time and risk preference measures. This approach allows us to capture potential non-linearities in the relationship between preferences and health behaviors, particularly in the form of diminishing marginal effects. The results, presented in Appendix 10 and 11, remained largely unchanged. The use of the log form did not meaningfully alter the magnitude or statistical significance of the estimates, and the direction of the effects was consistent with the linear specification.

These robustness tests suggest two possible explanations: (1) The statistical power of the combined dataset remains insufficient to detect the effect of preference because it is too small, or (2) Time preference and risk preference do not serve as mediating variables between education and health behavior.

5.5 Causal Mediation Analysis using *Mediate*

The results of the analysis using *mediate* command, summarized in Tables 5.15 through 5.20, show that while most estimates do not reach conventional levels of statistical significance, a consistent pattern emerges regarding smoking behavior. In both the IFLS-4 and IFLS-5 samples, increased year of education has monotonically positive effect on the probability of smoking.

These findings suggest that the relationship between education and smoking extends beyond financial considerations. Higher levels of education may increase exposure to social and professional environments where smoking is more common or socially acceptable. Moreover, the results highlight the heterogeneity of education's effect, as its impact on smoking behavior differs across levels of schooling.

Our results also show that in the IFLS-4 sample, education has a significant and positive overall effect on contraceptive use across all education levels. This effect remains robust even after controlling for household expenditure. In contrast, the IFLS-5 sample reveals a negative relationship between education and contraceptive use. We will discuss this discrepancy and reconcile it with the earlier IV-Probit results presented in Table 5.8 in the Discussion section.

Regarding health checkups, several of the estimated effects for individuals with early education become statistically significant once household expenditure is controlled for. Similar to the case of contraceptive use, this implies that the impact of education on health checkup behavior is also mediated through household expenditure.

VI. Discussion

Our findings have indicated that while, increased education due to SD Inpres program do affect all the health behaviors, risk preference and time preference are not meaningful mediators between education and health behavior. Instead, what we found was that economic level, as proxied by household expenditure, is the potential mediator in the relationship between the two variables.

While much of the existing literature has established that individual preferences, such as time and risk preferences, are significantly associated with various health behaviors including physical activity (Aswani et al., 2018), enrollment in health insurance (Tessema and Yismaw, 2023), smoking rates (Shahab and McEwen, 2009), and increased contraceptive use (Blank et al., 2023), studies that specifically examine whether these preferences mediate the relationship between education and health behaviors have found limited evidence of such a role. For instance, Takagi et al. (2016) in Japan and Cutler and Lleras-Muney (2010) in the United States both conclude that individual preferences account for only a very small fraction of the total effect of education on health behaviors. Our findings are consistent with these results, showing that while education significantly affects various health behaviors, risk and time preferences do not play a meaningful mediating role in this relationship.

In our analysis, economic level appears to be a more meaningful pathway linking education to health behaviors than individual preferences. This finding suggests that the benefits of education on health may operate not only through increased knowledge or behavioral traits but also through the material improvements that education facilitates. Higher education levels are likely to lead to better economic conditions, which in turn enable greater access to healthcare services, preventive behaviors, and healthier lifestyle choices. This interpretation is supported by previous research including Berkel et al. (2021) and Zhang et al.

(2024) which showed that material and economic conditions substantially mediate the relationship between education and health behaviors.

While Yamauchi and Saito (2019), using SUSENAS data, report that education reduces smoking prevalence among women, our analysis using IFLS data suggests otherwise. One key difference lies in the methodological approach: Yamauchi and Saito do not account for household expenditure or several individual-level characteristics that may confound the relationship, potentially biasing their estimates. In contrast, our model includes these controls and employs a decomposition approach using the IV-Probit and *mediate* command, allowing us to explore the effect across education levels. The result shows a consistently positive and increasing effect of education on smoking, particularly among women with higher levels of education. This pattern echoes the findings of Bratti, Cottini, and Ghinetti (2022), who document that higher education expansion in Italy was associated with increased smoking among women.

This finding, combined with the findings from IV-Probit, indicates the potentially opposing mechanisms through which education influences smoking behavior. On one hand, increased education raises household economic status, which has been documented to have negative effect on tobacco consumption in our study and previous ones (Hamad et al., 2018; Murakami et al., 2023; Pampel, 2003). On the other hand, exposure to education may also increase the likelihood of smoking among women through several social and behavioral channels.

First, education is associated with greater autonomy and shifts in gender norms, which can lead to the adoption of behaviors that were previously less socially acceptable for women, such as smoking (Hitchman and Fong, 2011; Nakkash et al., 2010). Second, more educated women are likely to be employed in urban or professional environments where smoking is more normalized and may function as a social or stress-coping behavior (Pampel, 2001; Boo

et al., 2022; Okechukwu et al., 2009). Third, higher education often leads to delayed marriage and childbearing, which may extend periods of individual autonomy and increase engagement in lifestyle choices such as smoking (Nagelhout et al., 2012; Brook et al., 2008; Graham et al., 2008). Fourth, although education typically reduces stress through improved socioeconomic conditions, it may also increase exposure to stressors such as job demands or gender role conflict, with smoking used as a coping strategy (Graham et al., 2006; Verplaetse et al., 2015; al'Absi et al., 2015). Finally, tobacco marketing has historically targeted educated, urban women by associating smoking with modernity, independence, and slimness, further reinforcing its appeal in these groups (Amos and Haglund, 2000; Brandt, 1996; Toll and Ling, 2005).

We find small evidence that time and risk preferences is significantly associated with tobacco use in the IFLS-5 sample. The relationship between time preference and smoking is marginally significant ($p = 0.098$; see Table 5.9), suggesting a weak tendency for more impatient individuals to be more likely to smoke. In contrast, the association between risk preference and smoking is statistically robust and remains significant even after controlling for household expenditure. Specifically, more risk-seeking individuals are significantly more likely to smoke, with the effect significant at the 1% level. These findings mimic results from previous studies which found that individuals with greater willingness to take risks are more likely to engage in smoking. Chetty et al. (2021) showed that risk-taking behavior is positively associated with smoking, while Van der Pol (2013) found that lower risk aversion contributes to higher smoking prevalence and explains part of the socioeconomic gradient in smoking.

For contraceptive use, we observe conflicting results across waves (see Table 5.10, Table 5.17, Table 5.18). While the IFLS-4 sample shows a positive effect of education, the IFLS-5 sample shows a negative effect. We argue that this discrepancy may be due to differences in the age range of respondents: 34–50 in IFLS-4 versus 43–57 in IFLS-5. Given

that the typical onset of menopause in Indonesia occurs around age 50 (Samil & Wishnuwardhani, 1994), many IFLS-5 respondents may be approaching or have already entered menopause. Since we lack direct data on menopausal status, it is possible that the observed decline in contraceptive use in IFLS-5 is driven more by biological factors than by educational attainment. In this context, we consider the IFLS-4 estimates to be more reliable for capturing the effect of education on contraceptive use among women who are still within their reproductive years.

The positive effect of education on contraceptive use found through the *mediate* result aligns with a large body of empirical literature. Numerous studies have shown that higher education is associated with increased contraceptive use and greater autonomy in reproductive decision-making. Bbaale and Mpuga (2011) found that secondary and higher education significantly increased modern contraceptive use among women in Uganda. Cleland et al. (2011) emphasized the importance of female education in driving demand for family planning services across developing countries, while Upadhyay and Karasek (2012) showed that education is positively associated with both contraceptive knowledge and use, even after controlling for wealth and urban status. These findings support the notion that education empowers women to make more informed reproductive choices.

That said, our findings should be interpreted within the broader understanding that the relationship between education and contraceptive use is unlikely to be uniform across all settings. While we observe a positive association, this effect is shaped by contextual factors such as cultural norms, gender roles, and social expectations. In some environments, increased education may not directly translate into greater contraceptive use if prevailing norms discourage autonomy in reproductive decisions. For instance, even educated women may face social pressures or partner opposition that constrain their ability to act on that knowledge. These complexities have been observed in other developing countries such as Pakistan and Uganda,

where the impact of education on fertility behavior is moderated by strong patriarchal structures and community-level norms (Choudhry et al., 2022; Nakphong et al., 2023).

In the IFLS-5 samples, after controlling for household expenditure, we observe a negative and significant association between years of education and the likelihood of undergoing a health checkup (Table 5.11). We also find supporting evidence from the *mediate* analysis in the IFLS-4 sample, although the significance level is lower (Table 5.19). Education may negatively impact the probability of undergoing health checkups by fostering an increased subjective perception of health. This explanation is supported by previous findings showing that individuals who perceive themselves as healthy are less likely to seek preventive healthcare services (Dryden et al., 2012; He et al., 2013; Noguchi and Shen, 2018).

To investigate this hypothesis, we estimated an IV-Probit model to assess the effect of education on subjective health perception. As shown in Appendix 8, the results indicate that additional years of education significantly increase the likelihood of individuals rating their health positively in the IFLS-5 sample. This finding supports our interpretation that education enhances subjective perceptions of health, which may in turn lower the probability of engaging in preventive health checkups.

We find that individuals with higher risk aversion (less risk seeking) are more likely to undergo health checkups suggests that personality traits may influence engagement with preventive care. This result is intuitive, those who are more cautious may be more inclined to reduce health-related uncertainty by seeking medical reassurance even in the absence of symptoms. It also points to the possibility that behavioral traits like risk preference can complement economic factors in shaping health decisions. While our data do not allow us to fully disentangle motivation from access, the result is consistent with patterns observed elsewhere, such as during the COVID-19 pandemic, when risk-averse individuals were more proactive in taking preventive measures (Lu et al., 2021). It also resonates with broader

evidence that risk aversion tends to increase the likelihood of using health services (Lutter et al., 2019). However, this effect diminishes after controlling for expenditure, lending support to our hypothesis that expenditure serves as a common mediator in both the education–health behaviors and preference–health behaviors relationships.

The observed relationship between risk preferences and health behaviors is primarily driven by responses from risk preference module B. This likely reflects the more extreme framing used in module B, where the hypothetical monetary gains and losses are large enough to deter most respondents from choosing the risky options. As a result, only individuals with particularly strong risk-seeking or risk-averse tendencies select the extreme choices. When combined with the low prevalence of reported health behaviors in the sample, this selective response pattern may contribute to the stronger and more significant association observed between module B risk preferences and health behaviors.

Our results indicate that while education has a significant effect on health behaviors, there is no significant evidence that individual preferences mediate this relationship. Instead, preferences appear to operate through separate pathways, potentially sharing a common mediator in economic status, as illustrated in the DAG in Appendix 13. This study has several limitations. First, the limited sample size reduces statistical power, especially for detecting effects related to preferences. Second, our method of eliciting preferences, although validated and commonly used in IFLS-based studies, is subject to well-known limitations of stated preference measures. The use of hypothetical binary choices may not accurately reflect real-world decision-making and is vulnerable to framing effects and measurement error, which may partly explain the weak association between preferences and health behaviors observed in our results. Third, we modelled time and risk preferences separately, whereas some frameworks consider them jointly, recognizing potential interactions between the two in shaping decision-making. Fourth, the time preference imputation method relies on the assumption of exponential

discounting, an assumption that has been shown to be violated in certain contexts. Fifth, we proxy for education using years of education, which may not fully capture the broader nuances of educational attainment, including differences in quality, field of study, pedagogy, and learning environment.

CHAPTER VII

CONCLUSION

This study finds that increased education from the SD Inpres program significantly affects health behaviors, with results showing both positive and negative impacts. Education is associated with higher contraceptive use, but also with increased smoking among women and a decrease in health checkup participation. These mixed findings reflect the complexity of how education interacts with individual decision-making.

Although both time and risk preferences are correlated with health behaviors, risk preferences show a stronger association than time preferences. However, the analysis finds no evidence that either preference mediates the effect of education. Instead, risk preference may operate through an independent pathway, while household expenditure appears to play a more significant role, suggesting that education influences health primarily through improvements in economic conditions.

This analysis simplifies education by using years of schooling as a proxy, which overlooks important nuances. Future research should explore how different dimensions of education, such as type, quality, and pedagogy, shape both individual preferences and health-related behaviors. A more comprehensive modelling approach may also be necessary to capture the complex interactions among education, preferences, and health outcomes. The use of larger datasets and more reliable methods for eliciting preferences, such as vignettes or real incentives, may help clarify the mechanisms underlying these relationships. Developing models that jointly incorporate risk and time preferences may also offer a more comprehensive understanding of how education shapes health outcomes.

While education shows mixed effects on specific health behaviors, its broader positive influence on health outcomes is well established. Policies that expand access to education

remain essential, but equal attention should be given to improving the quality of education, as this may affect how individuals translate knowledge into action. In addition, policies that aim to influence preferences directly may be another effective route to encouraging healthier behaviors.

Table 5.9 Smoking Behavior IV-Probit Analysis

Smoking Behavior											
IFLS-4											
Years of Education	0.217***	0.248***	0.225***	0.232***	0.228***	0.236***	0.236***	0.245***	0.237***	0.245***	0.243***
	(0.059)	(0.046)	(0.051)	(0.041)	(0.046)	(0.0438	(0.032)	(0.028)	(0.032)	(0.027)	(0.032)
Time Preference			-0.036	-0.062**							-0.066
Measure A			(0.044)	(0.035)							(0.042)
Time Preference					-0.290	-0.435					0.033
Measure B					(0.419)	(0.424)					(0.551)
Risk Preference							-0.008	-0.004			0.008
Measure A							(0.034)	(0.034)			(0.035)
Risk Preference									0.046	-0.043	-0.063
Measure B									(0.057)	(0.056)	(0.063)
Log(Expenditure)		-0.359**		-0.192**		-0.194***		-0.200***		-0.201**	-0.201***
		(0.036)		(0.079)		(0.075)		(0.063)		(0.062)	(0.067)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS-5											
Years of Education	0.217***	0.248***	0.220***	0.251***	0.222***	0.252***	0.226***	0.248***	0.211***	0.246***	0.248***
	(0.059)	(0.046)	(0.050)	(0.037)	(0.049)	(0.036)	(0.049)	(0.043)	(0.065)	(0.049)	(0.046)
Time Preference			-0.017	-0.014							0.009
Measure A			(0.023)	(0.021)							(0.023)
Time Preference					-0.528*	-0.494*					-0.448
Measure B					(0.310)	(0.282)					(0.331)
Risk Preference							0.000	-0.023			0.000
Measure A							(0.036)	(0.034)			(0.034)
Risk Preference									0.162***	0.162***	0.149***
Measure B									(0.060)	(0.053)	(0.056)
Log(Expenditure)		-0.359***		-0.364***		-0.364***		-0.359***		-0.358***	-0.358***
		(0.036)		(0.036)		(0.036)		(0.036)		(0.038)	(0.038)
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Table 5.10 Contraceptive Use IV-Probit Analysis

	Contraceptive Use										
	IFLS-4										
Years of Education	-0.053	-0.043	-0.055	-0.042	-0.053	-0.042	-0.055	-0.044	-0.053	-0.042	-0.042
	(0.276)	(0.215)	(0.277)	(0.212)	(0.273)	(0.213)	(0.276)	(0.216)	(0.275)	(0.215)	(0.212)
Time Preference			-0.004	0.002							0.002
Measure A			(0.040)	(0.041)							(0.048)
Time Preference					-0.013	-0.005					0.004
Measure B					(0.586)	(0.593)					(0.701)
Risk Preference							0.032	0.033			0.037
Measure A							(0.047)	(0.047)			(0.051)
Risk Preference									0.015	0.012	-0.015
Measure B									(0.095)	(0.089)	(0.105)
Log(Expenditure)		0.054		0.054		0.054		0.056		0.054	0.057
		(0.228)		(0.227)		(0.226)		(0.229)		(0.228)	(0.225)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057
	IFLS-5										
Years of Education	-0.244***	-0.260***	-0.244***	-0.260***	-0.245***	-0.261***	-0.243***	-0.261***	-0.243***	-0.260***	-0.261***
	(0.022)	(0.013)	(0.021)	(0.013)	(0.020)	(0.012)	(0.012)	(0.013)	(0.024)	(0.018)	(0.014)
Time Preference			-0.005	0.002							-0.029
Measure A			(0.034)	(0.035)							(0.050)
Time Preference					0.546	0.505					0.552
Measure B					(0.289)	(0.279)					(0.492)
Risk Preference							-0.032	-0.038			-0.018
Measure A							(0.031)	(0.031)			(0.030)
Risk Preference									-0.172***	-0.168***	-0.151***
Measure B									(0.054)	(0.054)	(0.055)
Log(Expenditure)		0.336***		0.335***		0.338***		0.338***		0.331***	0.337***
		(0.072)		(0.074)		(0.068)		(0.069)		(0.078)	(0.074)
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Table 5.11 Health Checkup IV-Probit Analysis

	Health Checkup										
	IFLS-4										
Years of Education	0.041	0.064	0.041	0.064	0.039	0.061	0.037	0.060	0.042	0.065	0.062
	(0.180)	(0.135)	(0.182)	(0.134)	(0.179)	(0.135)	(0.182)	(0.137)	(0.181)	(0.136)	(0.135)
Time Preference			-0.006	0.001							0.023
Measure A			(0.034)	(0.034)							(0.043)
Time Preference					-0.386	-0.384					-0.483
Measure B					(0.461)	(0.457)					(0.578)
Risk Preference							0.064	0.065			0.043
Measure A							(0.040)	(0.040)			(0.045)
Risk Preference									0.109	0.106	0.071
Measure B									(0.073)	(0.073)	(0.085)
Log(Expenditure)		0.059		0.059		0.060		0.061		0.057	0.061
		(0.143)		(0.142)		(0.142)		(0.143)		(0.143)	(0.142)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057
	IFLS-5										
Years of Education	-0.089	-0.237**	-0.105	-0.240***	-0.095	-0.240**	-0.082	-0.236***	-0.068	-0.230***	-0.232***
	(0.367)	(0.100)	(0.337)	(0.089)	(0.367)	(0.094)	(0.382)	(0.102)	(0.386)	(0.117)	(0.114)
Time Preference			0.049	0.028							0.011
Measure A			(0.044)	(0.044)							(0.058)
Time Preference					0.393	0.500*					0.261
Measure B					(0.624)	(0.288)					(0.506)
Risk Preference							0.007	-0.027			0.003
Measure A							(0.061)	(0.031)			(0.047)
Risk Preference									-0.240***	-0.222	-0.209
Measure B									(0.099)	(0.142)	(0.158)
Log(Expenditure)		0.418***		0.415***		0.415***		0.420***		0.429***	0.428***
		(0.127)		(0.127)		(0.132)		(0.126)		(0.117)	(0.122)
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Table 5.12 *Smoking Behavior* ivmediate Analysis

IFLS-4 Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)
Direct Effect	-0.001	-0.001	0.000	-0.000	-0.000	-0.001	0.011	0.013
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.166)	(0.196)
Indirect Effect	0.012	0.012	0.011	0.011	0.012	0.012	0.001	-0.002
	(0.025)	(0.019)	(0.028)	(0.024)	(0.050)	(0.019)	(0.172)	(0.169)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS-5 Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.104	0.040	0.104	0.040	0.104	0.040	0.104	0.040
	(0.352)	(0.081)	(0.352)	(0.081)	(0.352)	(0.081)	(0.352)	(0.081)
Direct Effect	-0.006	-0.007	-0.009	-0.008	-0.010	-0.007	-0.002	-0.002
	(0.007)	(0.007)	(0.010)	(0.010)	(0.031)	(0.026)	(0.004)	(0.004)
Indirect Effect	0.110	0.047	0.113	0.048	0.114	0.047	0.106	0.042
	(0.594)	(0.207)	(0.484)	(0.162)	(0.928)	(0.602)	(0.375)	(0.094)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Table 5.13 Contraceptive Use ivmediate Analysis

IFLS-4 Contraceptive Use								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.010	-0.007	-0.010	-0.007	-0.010	-0.007	-0.010	-0.007
	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)
Direct Effect	0.001	-0.001	-0.000	-0.001	-0.000	-0.001	-0.010	-0.008
	(0.003)	(0.002)	(0.003)	(0.004)	(0.003)	(0.004)	(0.149)	(0.107)
Indirect Effect	-0.011	-0.006	-0.010	-0.006	-0.010	-0.006	-0.000	0.001
	(0.040)	(0.028)	(0.039)	(0.027)	(0.052)	(0.032)	(0.149)	(0.087)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS-5 Contraceptive Use								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.253	-0.109	-0.253	-0.109	-0.253	-0.109	-0.253	-0.109
	(0.782)	(0.160)	(0.782)	(0.160)	(0.782)	(0.160)	(0.782)	(0.160)
Direct Effect	0.003	0.003	0.008	0.008	0.012	0.005	-0.007	-0.007
	(0.016)	(0.016)	(0.021)	(0.021)	(0.071)	(0.063)	(0.007)	(0.006)
Indirect Effect	-0.257	-0.113	-0.261	-0.117	-0.265	-0.114	-0.246	-0.103
	(1.327)	(0.471)	(1.095)	(0.373)	(2.134)	(1.448)	(0.829)	(0.194)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.14 Health Checkup ivmediate Analysis

IFLS-4 Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.016	0.02	0.016	0.018	0.016	0.018	0.016	0.018
	(0.41)	(0.58)	(0.41)	(0.58)	(0.41)	(0.58)	(0.41)	(0.58)
Direct Effect	0.017***	0.016***	0.017***	0.016***	0.017***	0.016***	0.016	0.018
	(5.69)	(5.01)	(4.55)	(3.65)	(5.50)	(5.01)	(0.39)	(0.40)
Indirect Effect	-0.001	0.001	-0.001	0.002	-0.001	0.002	-0.000	-0.000
	(-0.03)	(0.06)	(-0.03)	(0.06)	(-0.03)	(0.06)	(-0.00)	(-0.01)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS-5 Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.12	-0.03	-0.12	-0.03	-0.12	-0.03	-0.12	-0.03
	(0.41)	(0.09)	(0.41)	(0.09)	(0.41)	(0.09)	(0.41)	(0.09)
Direct Effect	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.01*
	(0.01)	(0.01)	(0.01)	(0.01)	(0.04)	(0.02)	(0.00)	(0.00)
Indirect Effect	-0.13	-0.04	-0.13	-0.04	-0.13	-0.04	-0.12	-0.04
	(0.70)	(0.20)	(0.56)	(0.15)	(1.06)	(0.53)	(0.44)	(0.09)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.15 IFLS-4 *Smoking Behavior* Mediate Analysis

IFLS-4 Smoking Behavior								
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	-0.001	-0.001	-0.000	-0.000	-0.001	-0.002	0.000	0.000
	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)
SMP	-0.002	-0.002	-0.000	-0.000	0.003	0.002	0.000	0.000
	(0.005)	(0.004)	(0.001)	(0.001)	(0.004)	(0.004)	(0.001)	(0.002)
SMA	-0.005	-0.004	0.000	0.000	0.017	0.015	-0.000	0.000
	(0.011)	(0.009)	(0.003)	(0.003)	(0.015)	(0.014)	(0.000)	(0.000)
Bachelor	-0.009	-0.008	0.002	0.002	0.061	0.057	-0.000	-0.000
	(0.022)	(0.020)	(0.010)	(0.009)	(0.046)	(0.046)	(0.005)	(0.003)
Natural Direct Effect								
SD	0.036**	0.032**	0.037**	0.033**	0.031**	0.027*	0.036**	0.032**
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.014)	(0.015)	(0.015)
SMP	0.078*	0.068*	0.082**	0.070*	0.066*	0.056	0.079*	0.068*
	(0.040)	(0.038)	(0.041)	(0.039)	(0.038)	(0.036)	(0.041)	(0.039)
SMA	0.143*	0.123	0.151*	0.129	0.121	0.102	0.146*	0.124
	(0.081)	(0.080)	(0.083)	(0.082)	(0.075)	(0.072)	(0.082)	(0.081)
Bachelor	0.268*	0.231	0.283*	0.244	0.224	0.191	0.274*	0.235
	(0.156)	(0.159)	(0.156)	(0.161)	(0.139)	(0.139)	(0.156)	(0.161)
Total Effect								
SD	0.035**	0.031**	0.037**	0.032**	0.029**	0.025*	0.036**	0.032**
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.014)	(0.015)	(0.014)
SMP	0.076*	0.066*	0.081**	0.070*	0.069*	0.058	0.079*	0.068*
	(0.040)	(0.038)	(0.041)	(0.039)	(0.039)	(0.037)	(0.041)	(0.039)
SMA	0.139*	0.119	0.151*	0.129	0.138*	0.117	0.146*	0.124
	(0.081)	(0.079)	(0.083)	(0.081)	(0.080)	(0.078)	(0.082)	(0.081)
Bachelor	0.259*	0.223	0.285*	0.246	0.285*	0.248	0.274*	0.235
	(0.157)	(0.159)	(0.155)	(0.160)	(0.154)	(0.159)	(0.157)	(0.161)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.16 IFLS-5 *Smoking Behavior* Mediate Analysis

IFLS-5 Smoking Behavior								
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	-0.001	-0.001	-0.003	-0.003	0.001	0.001	-0.001	-0.001
	(0.004)	(0.004)	(0.008)	(0.008)	(0.007)	(0.008)	(0.014)	(0.013)
SMP	0.000	0.000	-0.003	-0.003	-0.000	-0.001	-0.025	-0.024
	(0.010)	(0.010)	(0.016)	(0.017)	(0.004)	(0.006)	(0.042)	(0.041)
SMA	0.002	0.001	-0.000	-0.000	-0.002	-0.004	-0.044	-0.042
	(0.015)	(0.015)	(0.018)	(0.019)	(0.019)	(0.021)	(0.105)	(0.102)
Bachelor	0.005	0.004	0.004	0.004	-0.005	-0.010	-0.079	-0.077
	(0.029)	(0.029)	(0.035)	(0.037)	(0.049)	(0.055)	(0.176)	(0.172)
Natural Direct Effect								
SD	0.215*	0.213*	0.215*	0.214*	0.219**	0.217**	0.210*	0.209*
	(0.112)	(0.117)	(0.112)	(0.117)	(0.101)	(0.103)	(0.122)	(0.125)
SMP	0.423*	0.421*	0.424*	0.422*	0.438*	0.436*	0.432*	0.428*
	(0.239)	(0.250)	(0.237)	(0.248)	(0.228)	(0.231)	(0.244)	(0.250)
SMA	0.616***	0.614**	0.616***	0.615**	0.627***	0.628***	0.645***	0.641***
	(0.185)	(0.194)	(0.181)	(0.191)	(0.151)	(0.154)	(0.179)	(0.187)
Bachelor	0.831***	0.829***	0.830***	0.828***	0.849***	0.852***	0.886***	0.882***
	(0.204)	(0.212)	(0.200)	(0.209)	(0.169)	(0.167)	(0.137)	(0.145)
Total Effect								
SD	0.214*	0.213*	0.213*	0.211*	0.220**	0.219**	0.210*	0.208*
	(0.112)	(0.117)	(0.114)	(0.119)	(0.099)	(0.101)	(0.118)	(0.122)
SMP	0.423*	0.421*	0.421*	0.418*	0.437*	0.435*	0.407	0.404
	(0.237)	(0.248)	(0.239)	(0.251)	(0.228)	(0.231)	(0.252)	(0.257)
SMA	0.617***	0.616**	0.616***	0.614**	0.625***	0.624***	0.601**	0.599**
	(0.178)	(0.187)	(0.181)	(0.191)	(0.153)	(0.160)	(0.245)	(0.252)
Bachelor	0.836***	0.833***	0.835***	0.832***	0.844***	0.841***	0.807**	0.805**
	(0.189)	(0.198)	(0.191)	(0.200)	(0.174)	(0.181)	(0.262)	(0.267)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.17 IFLS-4 Contraceptive Use Mediate Analysis

	IFLS-4 Contraceptive Use							
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	-0.006	-0.006	-0.003	-0.003	0.002	0.002	0.000	0.000
	(0.009)	(0.010)	(0.005)	(0.006)	(0.004)	(0.005)	(0.004)	(0.004)
SMP	0.003	0.004	0.001	0.001	0.006	0.007	-0.000	-0.000
	(0.012)	(0.016)	(0.008)	(0.009)	(0.010)	(0.012)	(0.002)	(0.002)
SMA	0.024	0.027	0.012	0.011	0.011	0.013	-0.001	-0.002
	(0.030)	(0.033)	(0.021)	(0.021)	(0.021)	(0.023)	(0.018)	(0.018)
Bachelor	0.053	0.057	0.026	0.025	0.016	0.019	-0.003	-0.004
	(0.062)	(0.063)	(0.044)	(0.041)	(0.034)	(0.037)	(0.041)	(0.039)
Natural Direct Effect								
SD	0.137**	0.135**	0.127**	0.129**	0.122**	0.123**	0.128**	0.129**
	(0.061)	(0.063)	(0.060)	(0.062)	(0.060)	(0.062)	(0.061)	(0.062)
SMP	0.302**	0.300**	0.285**	0.291**	0.276**	0.280**	0.292**	0.296**
	(0.127)	(0.131)	(0.131)	(0.133)	(0.134)	(0.137)	(0.131)	(0.134)
SMA	0.489***	0.486***	0.474***	0.483***	0.464**	0.471***	0.490***	0.496***
	(0.160)	(0.162)	(0.174)	(0.170)	(0.184)	(0.182)	(0.171)	(0.167)
Bachelor	0.706***	0.700***	0.704***	0.710***	0.701***	0.704***	0.730***	0.734***
	(0.158)	(0.157)	(0.172)	(0.160)	(0.185)	(0.175)	(0.163)	(0.153)
Total Effect								
SD	0.131**	0.129**	0.124**	0.126**	0.124**	0.125**	0.128**	0.129**
	(0.061)	(0.063)	(0.059)	(0.061)	(0.060)	(0.062)	(0.061)	(0.063)
SMP	0.305**	0.304**	0.287**	0.292**	0.282**	0.287**	0.292**	0.296**
	(0.126)	(0.130)	(0.130)	(0.133)	(0.133)	(0.136)	(0.131)	(0.134)
SMA	0.513***	0.513***	0.486***	0.494***	0.476***	0.484***	0.488***	0.494***
	(0.150)	(0.150)	(0.170)	(0.165)	(0.178)	(0.174)	(0.169)	(0.167)
Bachelor	0.759***	0.757***	0.730***	0.735***	0.717***	0.723***	0.727***	0.730***
	(0.125)	(0.122)	(0.156)	(0.144)	(0.171)	(0.159)	(0.160)	(0.151)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.18 IFLS-5 Contraceptive Use Mediate Analysis

IFLS-5 Contraceptive Use								
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	-0.004	0.010	0.004	-0.010	-0.001	0.014	-0.005	0.005
	(0.012)	(0.015)	(0.009)	(0.014)	(0.009)	(0.012)	(0.009)	(0.008)
SMP	-0.007	0.016	0.002	-0.005	-0.001	0.010	-0.008	0.009
	(0.020)	(0.020)	(0.007)	(0.013)	(0.006)	(0.014)	(0.014)	(0.011)
SMA	-0.009	0.022	-0.001	0.003	-0.000	0.003	-0.010	0.012
	(0.027)	(0.030)	(0.006)	(0.016)	(0.002)	(0.019)	(0.018)	(0.017)
Bachelor	-0.005	0.027	-0.002	0.013	0.000	-0.007	-0.006	0.015
	(0.016)	(0.056)	(0.005)	(0.031)	(0.002)	(0.032)	(0.011)	(0.035)
Natural Direct Effect								
SD	-0.395***	-0.105	-0.402***	-0.106	-0.398***	-0.067	-0.394***	-0.095
	(0.022)	(0.330)	(0.018)	(0.336)	(0.017)	(0.290)	(0.021)	(0.321)
SMP	-0.564***	-0.144	-0.571***	-0.147	-0.570***	-0.094	-0.562***	-0.132
	(0.028)	(0.433)	(0.015)	(0.453)	(0.015)	(0.388)	(0.022)	(0.426)
SMA	-0.721***	-0.174	-0.726***	-0.181	-0.730***	-0.116	-0.718***	-0.161
	(0.054)	(0.497)	(0.050)	(0.531)	(0.043)	(0.459)	(0.054)	(0.495)
Bachelor	-0.829***	-0.202	-0.829***	-0.214	-0.836***	-0.139	-0.827***	-0.190
	(0.069)	(0.530)	(0.073)	(0.580)	(0.066)	(0.515)	(0.071)	(0.538)
Total Effect								
SD	-0.399***	-0.096	-0.397***	-0.115	-0.399***	-0.054	-0.399***	-0.090
	(0.017)	(0.324)	(0.019)	(0.341)	(0.016)	(0.286)	(0.017)	(0.320)
SMP	-0.572***	-0.128	-0.569***	-0.152	-0.571***	-0.083	-0.570***	-0.123
	(0.015)	(0.431)	(0.015)	(0.452)	(0.015)	(0.390)	(0.014)	(0.427)
SMA	-0.730***	-0.152	-0.727***	-0.178	-0.731***	-0.112	-0.729***	-0.149
	(0.045)	(0.507)	(0.049)	(0.529)	(0.043)	(0.460)	(0.046)	(0.503)
Bachelor	-0.834***	-0.175	-0.831***	-0.201	-0.835***	-0.146	-0.833***	-0.175
	(0.068)	(0.574)	(0.073)	(0.597)	(0.066)	(0.499)	(0.069)	(0.565)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.19 IFLS-4 Health Check Mediate Analysis

IFLS-4 Health Check								
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	-0.003	-0.004	0.004	0.001	0.004	0.002	0.002	-0.002
	(0.008)	(0.008)	(0.009)	(0.004)	(0.011)	(0.008)	(0.005)	(0.005)
SMP	0.007	0.005	0.007	0.002	0.011	0.006	0.006	-0.005
	(0.013)	(0.008)	(0.018)	(0.007)	(0.015)	(0.009)	(0.011)	(0.007)
SMA	0.025	0.015	0.012	0.002	0.021	0.010	0.011	-0.007
	(0.038)	(0.013)	(0.031)	(0.009)	(0.029)	(0.012)	(0.022)	(0.011)
Bachelor	0.064	0.024	0.020	0.003	0.040	0.012	0.022	-0.008
	(0.119)	(0.022)	(0.057)	(0.010)	(0.072)	(0.015)	(0.050)	(0.014)
Natural Direct Effect								
SD	-0.019	-0.210*	-0.002	-0.193	-0.021	-0.208*	0.000	-0.183
	(0.236)	(0.115)	(0.213)	(0.122)	(0.238)	(0.114)	(0.210)	(0.125)
SMP	-0.027	-0.297*	-0.003	-0.271	-0.031	-0.293*	0.001	-0.258
	(0.346)	(0.168)	(0.318)	(0.174)	(0.345)	(0.165)	(0.316)	(0.178)
SMA	-0.034	-0.365*	-0.004	-0.333	-0.041	-0.360*	0.001	-0.318
	(0.451)	(0.205)	(0.422)	(0.211)	(0.444)	(0.201)	(0.423)	(0.216)
Bachelor	-0.041	-0.424*	-0.005	-0.391*	-0.053	-0.420*	0.002	-0.376
	(0.582)	(0.226)	(0.560)	(0.236)	(0.563)	(0.223)	(0.565)	(0.243)
Total Effect								
SD	-0.022	-0.214*	0.001	-0.192	-0.018	-0.206*	0.003	-0.186
	(0.237)	(0.116)	(0.211)	(0.122)	(0.238)	(0.115)	(0.209)	(0.126)
SMP	-0.020	-0.293*	0.004	-0.269	-0.021	-0.287*	0.007	-0.263
	(0.346)	(0.168)	(0.319)	(0.174)	(0.346)	(0.165)	(0.317)	(0.177)
SMA	-0.009	-0.350*	0.008	-0.330	-0.020	-0.350*	0.013	-0.325
	(0.464)	(0.206)	(0.429)	(0.211)	(0.455)	(0.202)	(0.430)	(0.214)
Bachelor	0.023	-0.400*	0.015	-0.388	-0.013	-0.408*	0.024	-0.384
	(0.657)	(0.236)	(0.584)	(0.236)	(0.609)	(0.227)	(0.592)	(0.238)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Table 5.20 IFLS-5 Health Check Mediate Analysis

	TIFLS-5 Health Checkup							
	Time Preference				Risk Preference			
	Measure A		Measure B		Measure A		Measure B	
Natural Indirect Effect								
SD	0.001	-0.001	-0.001	0.000	-0.000	-0.000	-0.045	-0.016
	(0.006)	(0.004)	(0.008)	(0.005)	(0.004)	(0.003)	(0.093)	(0.028)
SMP	0.011	-0.007	0.004	-0.003	-0.002	-0.002	-0.116	-0.032
	(0.036)	(0.014)	(0.021)	(0.011)	(0.021)	(0.007)	(0.197)	(0.050)
SMA	0.037	-0.022	0.021	-0.012	-0.006	-0.007	-0.213	-0.057
	(0.114)	(0.045)	(0.058)	(0.029)	(0.060)	(0.020)	(0.374)	(0.101)
Bachelor	0.079	-0.060	0.054	-0.037	-0.013	-0.019	-0.309	-0.100
	(0.243)	(0.128)	(0.135)	(0.082)	(0.123)	(0.055)	(0.602)	(0.207)
Natural Direct Effect								
SD	0.060	0.042	0.061	0.041	0.062	0.041	0.090	0.049
	(0.068)	(0.057)	(0.067)	(0.052)	(0.060)	(0.051)	(0.072)	(0.079)
SMP	0.157	0.090	0.157	0.084	0.165	0.083	0.219	0.091
	(0.207)	(0.134)	(0.206)	(0.126)	(0.193)	(0.123)	(0.241)	(0.160)
SMA	0.309	0.162	0.310	0.150	0.332	0.145	0.408	0.148
	(0.399)	(0.274)	(0.400)	(0.260)	(0.378)	(0.251)	(0.476)	(0.291)
Bachelor	0.555	0.299	0.560	0.273	0.605	0.260	0.684	0.247
	(0.641)	(0.546)	(0.638)	(0.528)	(0.589)	(0.507)	(0.686)	(0.538)
Total Effect								
SD	0.062	0.041	0.060	0.041	0.062	0.040	0.045	0.033
	(0.069)	(0.058)	(0.067)	(0.052)	(0.061)	(0.051)	(0.076)	(0.088)
SMP	0.168	0.083	0.161	0.081	0.163	0.080	0.102	0.058
	(0.212)	(0.132)	(0.209)	(0.126)	(0.193)	(0.122)	(0.188)	(0.153)
SMA	0.346	0.141	0.330	0.138	0.326	0.138	0.195	0.091
	(0.410)	(0.258)	(0.408)	(0.252)	(0.378)	(0.246)	(0.389)	(0.249)
Bachelor	0.634	0.239	0.613	0.236	0.592	0.242	0.375	0.146
	(0.631)	(0.499)	(0.630)	(0.494)	(0.592)	(0.490)	(0.750)	(0.437)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

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APPENDIX

Appendix 1. Placebo Test Result

	Years of Education	
	IFLS-4	IFLS-5
INPRES Intensity * Treated Cohort Dummy	-0.007 (0.989)	0.153 (0.202)
Observations	1,057	1,114
F-Statistics	0.576	1.218

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency

Appendix 2. Smoking Behavior IV-Probit Average Marginal Effect

Smoking Behavior											
IFLS-4											
Years of	0.047	0.044	0.036	0.034	0.039	0.037	0.038	0.036	0.039	0.037	0.039
Education	(0.050)	(0.033)	(0.040)	(0.026)	(0.043)	(0.029)	(0.042)	(0.028)	(0.043)	(0.029)	(0.029)
Time Preference			-0.004	-0.008							-0.010
Measure A			(0.005)	(0.006)							(0.007)
Time Preference					-0.045	-0.065					0.002
Measure B					(0.084)	(0.080)					(0.084)
Risk Preference							-0.002	-0.001			0.003
Measure A							(0.006)	(0.005)			(0.006)
Risk Preference									-0.005	-0.004	-0.010
Measure B									(0.012)	(0.010)	(0.012)
Log(Expenditure)		-0.038		-0.029		-0.032		-0.031		-0.032	-0.034
		(0.034)		(0.028)		(0.031)		(0.030)		(0.030)	(0.031)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

Smoking Behavior											
IFLS-5											
Years of	0.057	0.127	0.067	0.156	0.068	0.165	0.067	0.158	0.061	0.165	0.114
Education	(0.090)	(0.356)	(0.099)	(0.456)	(0.104)	(0.506)	(0.099)	(0.452)	(0.085)	(0.506)	(0.290)
Time Preference			-0.005	-0.008							0.005
Measure A			(0.009)	(0.024)							(0.015)
Time Preference					-0.159	-0.317					0.002
Measure B					(0.204)	(0.875)					(0.084)
Risk Preference							0.000	0.014			-0.205
Measure A							(0.010)	(0.053)			(0.448)
Risk Preference									0.042	0.081	-0.010
Measure B									(0.053)	(0.018)	(0.012)
Log(Expenditure)		-0.183		-0.225		-0.237		-0.227		-0.188	-0.164
		(0.485)		(0.621)		(0.688)		(0.619)		(0.436)	(0.394)
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Appendix 3. Contraceptive Use IV-Probit Average Marginal Effect

	Contraceptive Use										
	IFLS-4										
Years of	-0.008	-0.007	-0.008	-0.006	-0.008	-0.007	-0.009	-0.007	-0.008	-0.006	-0.007
Education	(0.044)	(0.034)	(0.045)	(0.033)	(0.044)	(0.033)	(0.045)	(0.034)	(0.044)	(0.034)	(0.034)
Time Preference			-0.001	0.000							0.000
Measure A			(0.006)	(0.006)							(0.007)
Time Preference					-0.002	-0.001					0.000
Measure B					(0.090)	(0.091)					(0.107)
Risk Preference							0.005	0.005			0.005
Measure A							(0.007)	(0.007)			(0.008)
Risk Preference									0.002	0.002	-0.002
Measure B									(0.015)	(0.014)	(0.016)
Log(Expenditure)		0.008		0.008		0.008		0.009		0.008	0.008
		(0.036)		(0.036)		(0.036)		(0.036)		(0.036)	(0.036)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

	Contraceptive Use										
	IFLS-5										
Years of	-0.166	-0.340	-0.167	-0.344	-0.173	-0.369	-0.167	-0.343	-0.149	-0.279	-0.302
Education	(0.228)	(0.931)	(0.231)	(0.939)	(0.244)	(1.015)	(0.230)	(0.931)	(0.193)	(0.673)	(0.746)
Time Preference			-0.003	0.003							-0.033
Measure A			(0.020)	(0.051)							(0.038)
Time Preference					0.378	0.699					0.629
Measure B					(0.480)	(1.791)					(1.136)
Risk Preference							-0.020	-0.047			-0.018
Measure A							(0.023)	(0.109)			(0.041)
Risk Preference									0.105	0.178	-0.175
Measure B									(0.120)	(0.395)	(0.399)
Log(Expenditure)		0.437		0.441		0.476		0.445		0.353	0.385
		(1.258)		(1.268)		(1.371)		(1.265)		(0.906)	(1.006)
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

Appendix 4. Health Checkup IV-Probit Average Marginal Effect

	Health Checkup										
	IFLS-4										
Years of Education	0.009 (0.037)	0.013 (0.028)	0.009 (0.038)	0.013 (0.028)	0.008 (0.037)	0.013 (0.028)	0.008 (0.038)	0.013 (0.028)	0.009 (0.037)	0.013 (0.028)	0.013 (0.028)
Time Preference Measure A			-0.001 (0.007)	0.000 (0.007)							0.004 (0.009)
Time Preference Measure B					-0.082 (0.095)	-0.080 (0.095)					-0.100 (0.119)
Risk Preference Measure A							0.014 (0.008)	0.014 (0.008)			0.009 (0.009)
Risk Preference Measure B									0.023 (0.016)	0.022 (0.015)	0.014 (0.017)
Log(Expenditure)		0.012 (0.030)		0.012 (0.030)		0.012 (0.030)		0.013 (0.030)		0.012 (0.030)	0.012 (0.030)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

	Health Checkup										
	IFLS-5										
Years of Education	-0.019 (0.076)	-0.116 (0.367)	-0.022 (0.079)	-0.126 (0.398)	-0.020 (0.079)	-0.127 (0.422)	-0.017 (0.075)	-0.114 (0.361)	-0.014 (0.068)	-0.092 (0.260)	-0.097 (0.286)
Time Preference Measure A			0.008 (0.007)	0.013 (0.022)							0.003 (0.015)
Time Preference Measure B					0.069 (0.159)	0.252 (0.745)					0.111 (0.440)
Risk Preference Measure A							0.001 (0.008)	-0.013 (0.042)			0.001 (0.015)
Risk Preference Measure B									0.037 (0.045)	0.084 (0.154)	-0.082 (0.154)
Log(Expenditure)		0.198 (0.497)		0.210 (0.538)		0.211 (0.571)		0.195 (0.491)		0.165 (0.351)	0.172 (0.386)
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, chronic disease, and number of dependencies.

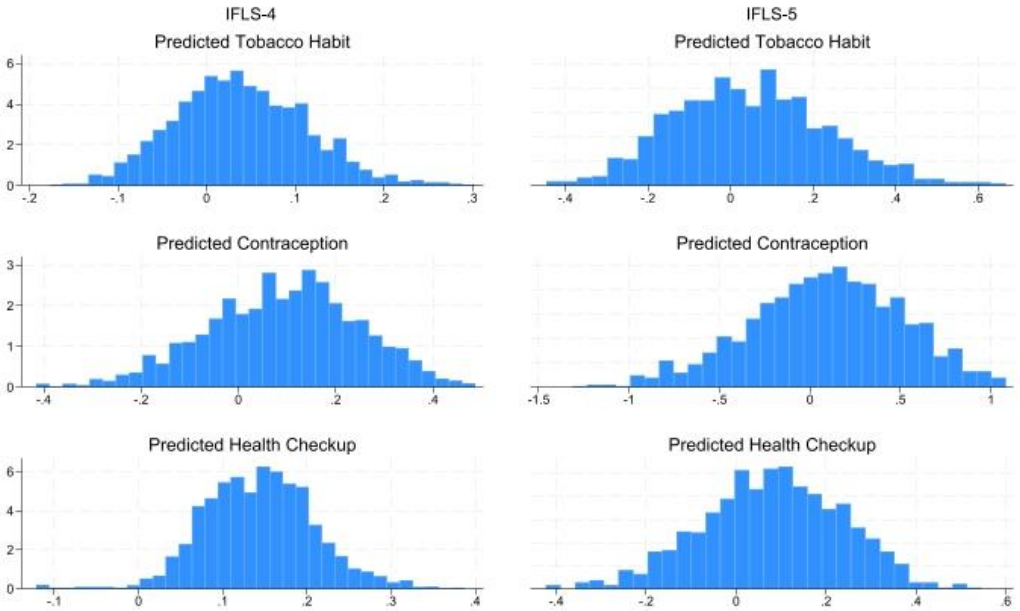
Appendix 5. Weekly Tobacco Expenditure ivmediate Analysis

IFLS-4 Tobacco Expenditure								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-52.31	-11.51	-52.31	-11.51	-52.31	-11.51	-52.31	-11.51
	(659.58)	(515.80)	(659.58)	(515.80)	(659.58)	(515.80)	(659.58)	(515.80)
Direct Effect	14.97	-2.76	8.90	-3.36	10.90	-3.23	-49.37	-13.01
	(60.68)	(64.69)	(73.74)	(84.49)	(63.42)	(79.14)	(1059.47)	(604.24)
Indirect Effect	-67.27	-8.75	-61.21	-8.15	-63.21	-8.28	-2.94	1.50
	(679.97)	(485.47)	(630.90)	(452.84)	(682.63)	(460.24)	(936.77)	(147.42)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS-5 Tobacco Expenditure								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	3061.1	5968.9	3061.1	5968.9	3061.1	5968.9	3061.1	5968.9
	(3212.3)	(7657.7)	(3212.3)	(7657.7)	(3212.3)	(7657.7)	(3212.3)	(7657.7)
Direct Effect	-378.0	-415.7	-541.9	-571.9	-1011.1	-1021.8	82.2	85.9
	(1719.5)	(1751.7)	(1444.2)	(1383.7)	(15511.6)	(5154.9)	(218.9)	(242.1)
Indirect Effect	3439.1	6384.7	3603.0	6540.9	4072.2	6990.8	2978.8	5883.0
	(32767)	(60731)	(13580)	(25970)	(90531)	(55416)	(3411)	(8616)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 6. Distribution of Predicted Values



Appendix 7. Pooled Dataset *ivmediate* Analysis

Pooled Data Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
	(0.024)	(0.025)	(0.024)	(0.025)	(0.024)	(0.025)	(0.024)	(0.025)
Direct Effect	-0.004	-0.003	-0.004	-0.005	-0.002	-0.001	-0.003	-0.003
	(0.006)	(0.008)	(0.002)	(0.004)	(0.010)	(0.016)	(0.003)	(0.003)
Indirect Effect	0.009	0.008	0.009	0.010	0.006	0.006	0.008	0.008
	(0.083)	(0.065)	(0.041)	(0.050)	(0.065)	(0.058)	(0.022)	(0.024)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,684	1,684	1,684	1,684	1,684	1,684	1,684	1,684

Pooled Data Contraceptive								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.019	-0.020	-0.019	-0.020	-0.019	-0.020	-0.019	-0.020
	(0.024)	(0.025)	(0.024)	(0.025)	(0.024)	(0.025)	(0.024)	(0.025)
Direct Effect	0.002	-0.003	0.001	0.002	-0.004	-0.007	-0.002	-0.002
	(0.011)	(0.012)	(0.005)	(0.007)	(0.026)	(0.041)	(0.003)	(0.002)
Indirect Effect	-0.021	-0.017	-0.020	-0.022	-0.015	-0.012	-0.017	-0.018
	(0.155)	(0.115)	(0.065)	(0.076)	(0.154)	(0.130)	(0.033)	(0.035)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,684	1,684	1,684	1,684	1,684	1,684	1,684	1,684

Pooled Data Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014
	(0.012)	(0.013)	(0.012)	(0.013)	(0.012)	(0.013)	(0.012)	(0.013)
Direct Effect	0.010***	0.010***	0.010***	0.009***	0.011**	0.011	0.011***	0.010***
	(0.003)	(0.004)	(0.002)	(0.003)	(0.005)	(0.009)	(0.002)	(0.002)
Indirect Effect	0.004	0.004	0.004	0.005	0.003	0.003	0.003	0.004
	(0.030)	(0.028)	(0.019)	(0.025)	(0.027)	(0.027)	(0.011)	(0.013)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,684	1,684	1,684	1,684	1,684	1,684	1,684	1,684

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 8. IFLS4 Binary *ivmediate* Analysis

IFLS4 Binary Preference Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)
Direct Effect	-0.002	-0.002	-0.001	-0.001	-0.016	-0.001	0.000	0.001
	(0.002)	(0.003)	(0.002)	(0.002)	(0.630)	(0.014)	(0.007)	(0.005)
Indirect Effect	0.013	0.013	0.012	0.012	0.027	0.013	0.011	0.011
	(0.021)	(0.028)	(0.023)	(0.028)	(1.290)	(0.223)	(0.045)	(0.047)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS4 Binary Preference Contraceptive								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.007	-0.010	-0.007	-0.010	-0.007	-0.010	-0.007	-0.010
	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)
Direct Effect	-0.000	0.001	-0.001	0.000	0.007	0.001	-0.001	-0.001
	(0.002)	(0.004)	(0.002)	(0.002)	(0.329)	(0.012)	(0.006)	(0.006)
Indirect Effect	-0.007	-0.011	-0.006	-0.011	-0.014	-0.011	-0.005	-0.009
	(0.029)	(0.042)	(0.029)	(0.041)	(0.672)	(0.199)	(0.033)	(0.051)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

IFLS4 Binary Preference Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.018	0.016	0.018	0.016	0.018	0.016	0.018	0.016
	(0.031)	(0.040)	(0.031)	(0.040)	(0.031)	(0.040)	(0.031)	(0.040)
Direct Effect	0.016***	0.018***	0.016***	0.018***	0.014	0.018***	0.017***	0.017***
	(0.003)	(0.005)	(0.003)	(0.003)	(0.095)	(0.004)	(0.006)	(0.006)
Indirect Effect	0.002	-0.001	0.002	-0.001	0.004	-0.001	0.002	-0.001
	(0.031)	(0.043)	(0.030)	(0.041)	(0.193)	(0.049)	(0.027)	(0.035)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,057	1,057	1,057	1,057	1,057	1,057	1,057	1,057

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 9. IFLS5 Binary Preference *ivmediate* Analysis

IFLS5 Binary Preference Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.104	0.040	0.104	0.040	0.104	0.040	0.104	0.040
	(0.352)	(0.081)	(0.352)	(0.081)	(0.352)	(0.081)	(0.352)	(0.081)
Direct Effect	-0.003	-0.003	-0.005**	-0.005**	-0.006**	-0.005**	0.015	0.011
	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.002)	(0.107)	(0.073)
Indirect Effect	0.107	0.043	0.109	0.045	0.110	0.045	0.089	0.029
	(0.469)	(0.142)	(0.434)	(0.130)	(0.409)	(0.114)	(0.826)	(0.254)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS5 Binary Preference Contraceptive								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.253	-0.109	-0.253	-0.109	-0.253	-0.109	-0.253	-0.109
	(0.782)	(0.160)	(0.782)	(0.160)	(0.782)	(0.160)	(0.782)	(0.160)
Direct Effect	-0.005	-0.006	-0.001	0.000	0.002	-0.000	-0.047	-0.039
	(0.009)	(0.009)	(0.005)	(0.006)	(0.005)	(0.004)	(0.236)	(0.166)
Indirect Effect	-0.248	-0.104	-0.253	-0.110	-0.255	-0.109	-0.206	-0.071
	(1.048)	(0.317)	(0.973)	(0.291)	(0.913)	(0.247)	(1.888)	(0.604)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS5 Binary Preference Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.119	-0.030	-0.119	-0.030	-0.119	-0.030	-0.119	-0.030
	(0.415)	(0.087)	(0.415)	(0.087)	(0.415)	(0.087)	(0.415)	(0.087)
Direct Effect	0.005	0.008	0.007**	0.010***	0.008**	0.010***	-0.016	-0.004
	(0.005)	(0.005)	(0.003)	(0.003)	(0.004)	(0.003)	(0.122)	(0.066)
Indirect Effect	-0.123	-0.038	-0.126	-0.040	-0.127	-0.040	-0.103	-0.026
	(0.550)	(0.138)	(0.503)	(0.125)	(0.471)	(0.110)	(0.951)	(0.227)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 10. IFLS4 Non-linear Preferences *ivmediate* Analysis

IFLS4 Non-linear Preference Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)
Direct Effect	-0.001	-0.001	-0.000	0.000	-0.001	-0.001	0.015	0.013
	(0.002)	(0.001)	(0.003)	(0.002)	(0.002)	(0.002)	(0.272)	(0.233)
Indirect Effect	0.012	0.012	0.011	0.011	0.012	0.012	-0.005	-0.002
	(0.019)	(0.025)	(0.024)	(0.028)	(0.029)	(0.032)	(0.211)	(0.207)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS4 Non-linear Preference Contraceptive								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.007	-0.010	-0.007	-0.010	-0.007	-0.010	-0.007	-0.010
	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)	(0.028)	(0.037)
Direct Effect	-0.001	0.000	-0.001	-0.001	-0.001	0.000	-0.009	-0.011
	(0.003)	(0.002)	(0.004)	(0.004)	(0.002)	(0.002)	(0.146)	(0.206)
Indirect Effect	-0.006	-0.011	-0.006	-0.010	-0.006	-0.010	0.002	0.001
	(0.028)	(0.039)	(0.027)	(0.039)	(0.030)	(0.042)	(0.109)	(0.179)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS4 Non-linear Preference Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.018	0.016	0.018	0.016	0.018	0.016	0.018	0.016
	(0.031)	(0.040)	(0.031)	(0.040)	(0.031)	(0.040)	(0.031)	(0.040)
Direct Effect	0.017***	0.017***	0.017***	0.017***	0.016***	0.017***	0.019	0.016
	(0.004)	(0.003)	(0.005)	(0.004)	(0.003)	(0.003)	(0.059)	(0.050)
Indirect Effect	0.002	-0.001	0.002	-0.001	0.002	-0.001	-0.001	0.000
	(0.029)	(0.040)	(0.027)	(0.037)	(0.030)	(0.040)	(0.033)	(0.023)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 11. IFLS5 Non-linear Preferences *ivmediate* Analysis

IFLS5 Non-linear Preference Smoking Behavior								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	0.095	0.038	0.095	0.038	0.095	0.038	0.095	0.038
	(0.295)	(0.074)	(0.295)	(0.074)	(0.295)	(0.074)	(0.295)	(0.074)
Direct Effect	-0.010	-0.011	-0.008	-0.008	-0.168	0.003	-0.002	-0.003
	(0.031)	(0.029)	(0.009)	(0.008)	(19.547)	(0.103)	(0.003)	(0.003)
Indirect Effect	0.106	0.048	0.104	0.046	0.263	0.035	0.098	0.041
	(0.775)	(0.305)	(0.395)	(0.136)	(33.680)	(0.716)	(0.315)	(0.085)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS5 Non-linear Preference Contraceptive								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.216	-0.098	-0.216	-0.098	-0.216	-0.098	-0.216	-0.098
	(0.605)	(0.137)	(0.605)	(0.137)	(0.605)	(0.137)	(0.605)	(0.137)
Direct Effect	0.011	0.011	0.007	0.006	0.349	-0.020	-0.006	-0.006
	(0.062)	(0.062)	(0.016)	(0.017)	(41.982)	(0.233)	(0.005)	(0.005)
Indirect Effect	-0.227	-0.110	-0.223	-0.105	-0.565	-0.079	-0.210	-0.093
	(1.618)	(0.666)	(0.823)	(0.291)	(72.337)	(1.628)	(0.641)	(0.164)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

IFLS5 Non-linear Preference Health Checkup								
	Time Preference				Risk Preference			
	Module A		Module B		Module A		Module B	
Total Effect	-0.119	-0.030	-0.119	-0.030	-0.119	-0.030	-0.119	-0.030
	(0.415)	(0.087)	(0.415)	(0.087)	(0.415)	(0.087)	(0.415)	(0.087)
Direct Effect	0.005	0.008	0.007**	0.010***	0.008**	0.010***	-0.016	-0.004
	(0.005)	(0.005)	(0.003)	(0.003)	(0.004)	(0.003)	(0.122)	(0.066)
Indirect Effect	-0.123	-0.038	-0.126	-0.040	-0.127	-0.040	-0.103	-0.026
	(0.550)	(0.138)	(0.503)	(0.125)	(0.471)	(0.110)	(0.951)	(0.227)
Expenditure Control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,144	1,144	1,144	1,144	1,144	1,144	1,144	1,144

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. Higher values of the time preference measure indicate greater impatience. Higher values of the risk preference indicate greater risk seeking. The regression includes controls for year of birth interacted with the population aged 5 to 14 in 1971, district enrollment rate in 1971, allocation of the water and sanitation program, and individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 12. Health Perception IV-Probit

	Rating One's Condition as Healthy	
	IFLS-4	IFLS-5
Education	0.161**	0.235***
	(0.108)	(0.045)
Observations	1,057	1,114

NOTE: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors are reported in parentheses. The regression includes controls for individual-level covariates including a dummy for Islamic religion, Javanese ethnicity, language, age, urban-rural, and dependency.

Appendix 13. Suggested DAG Graph

